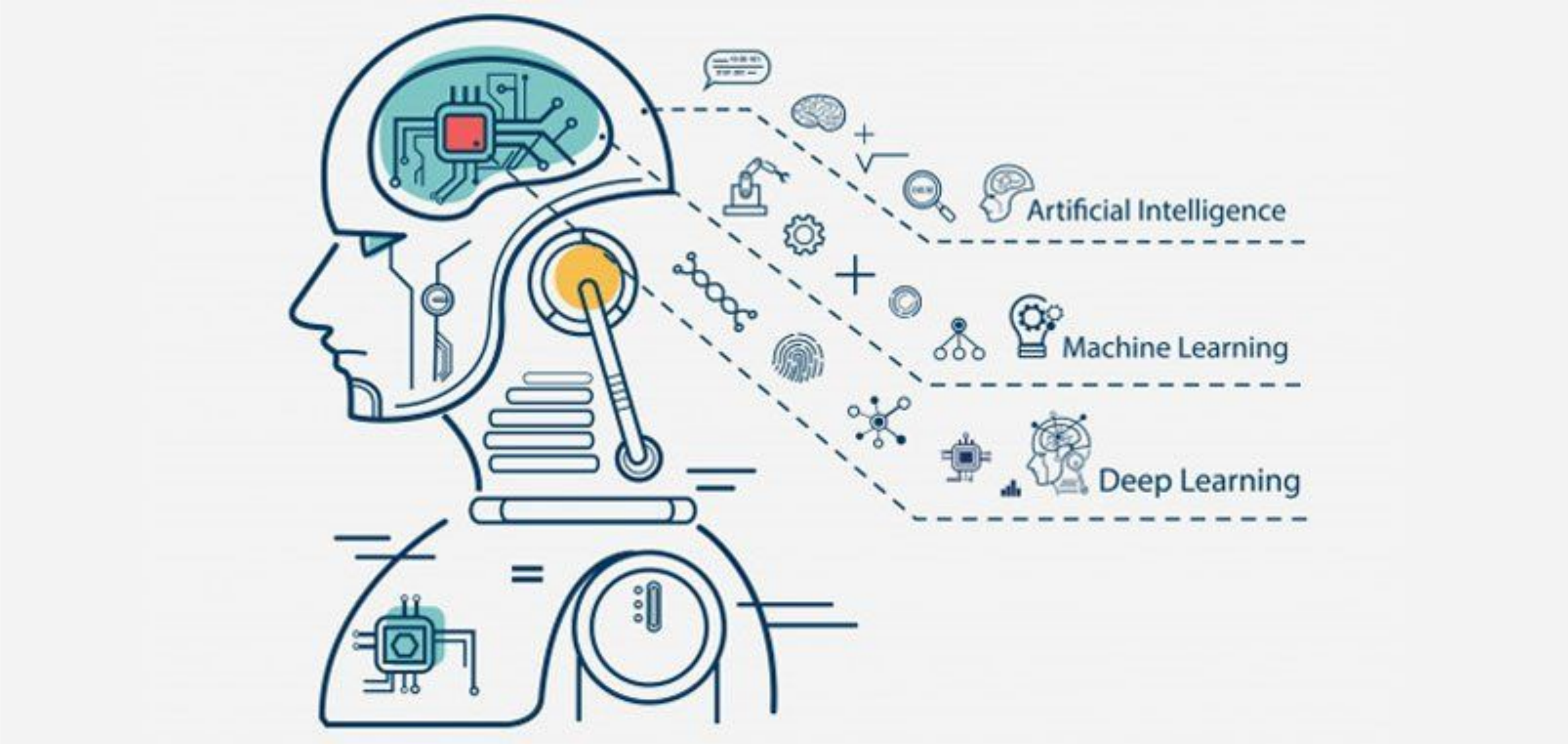
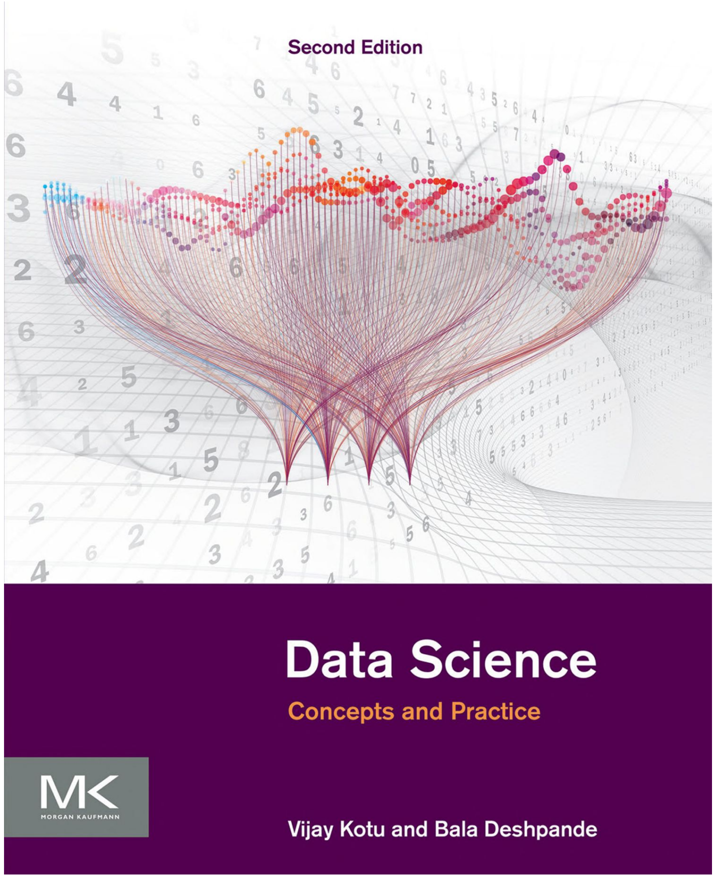
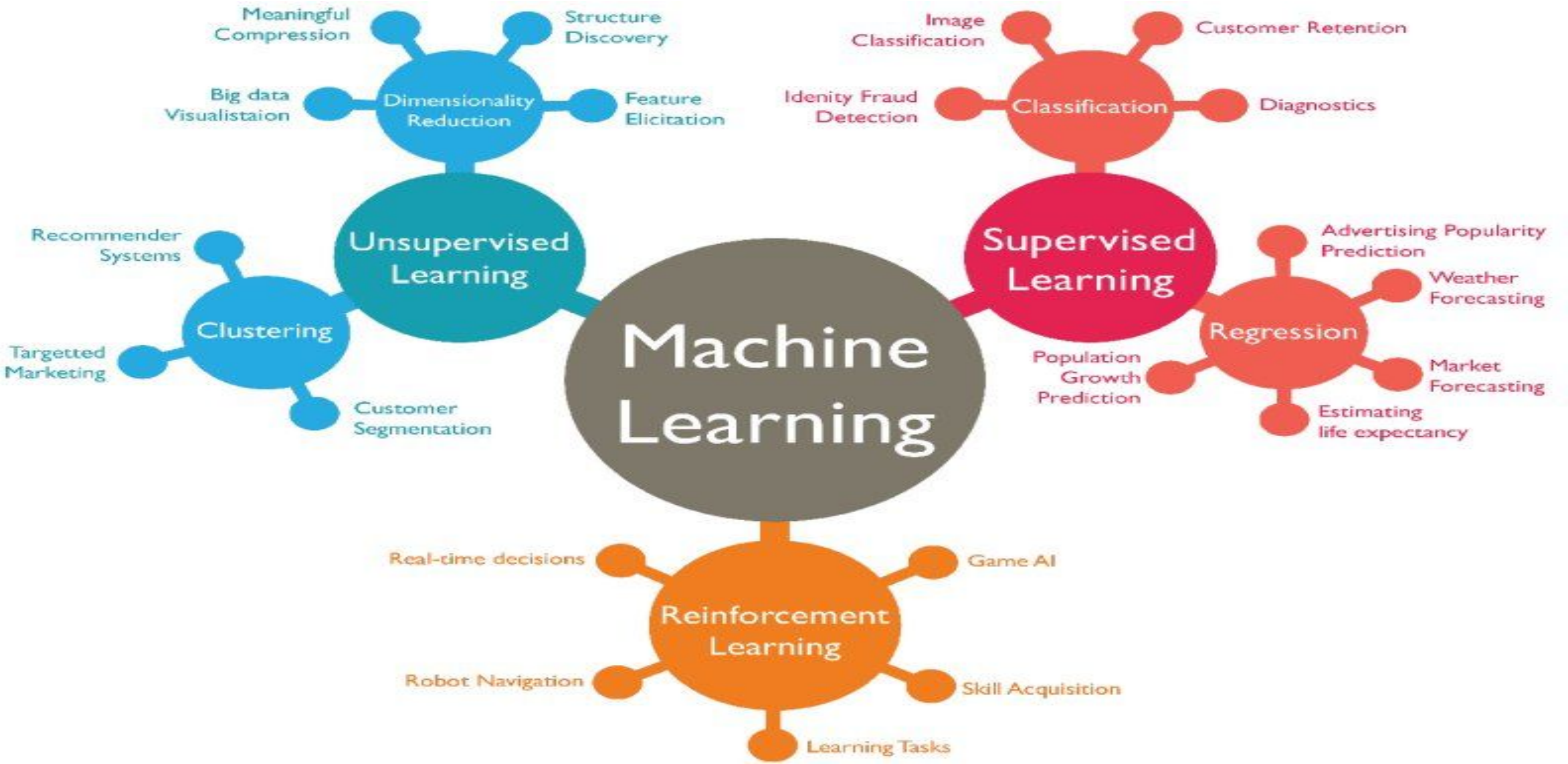
Data science & Machine Learning



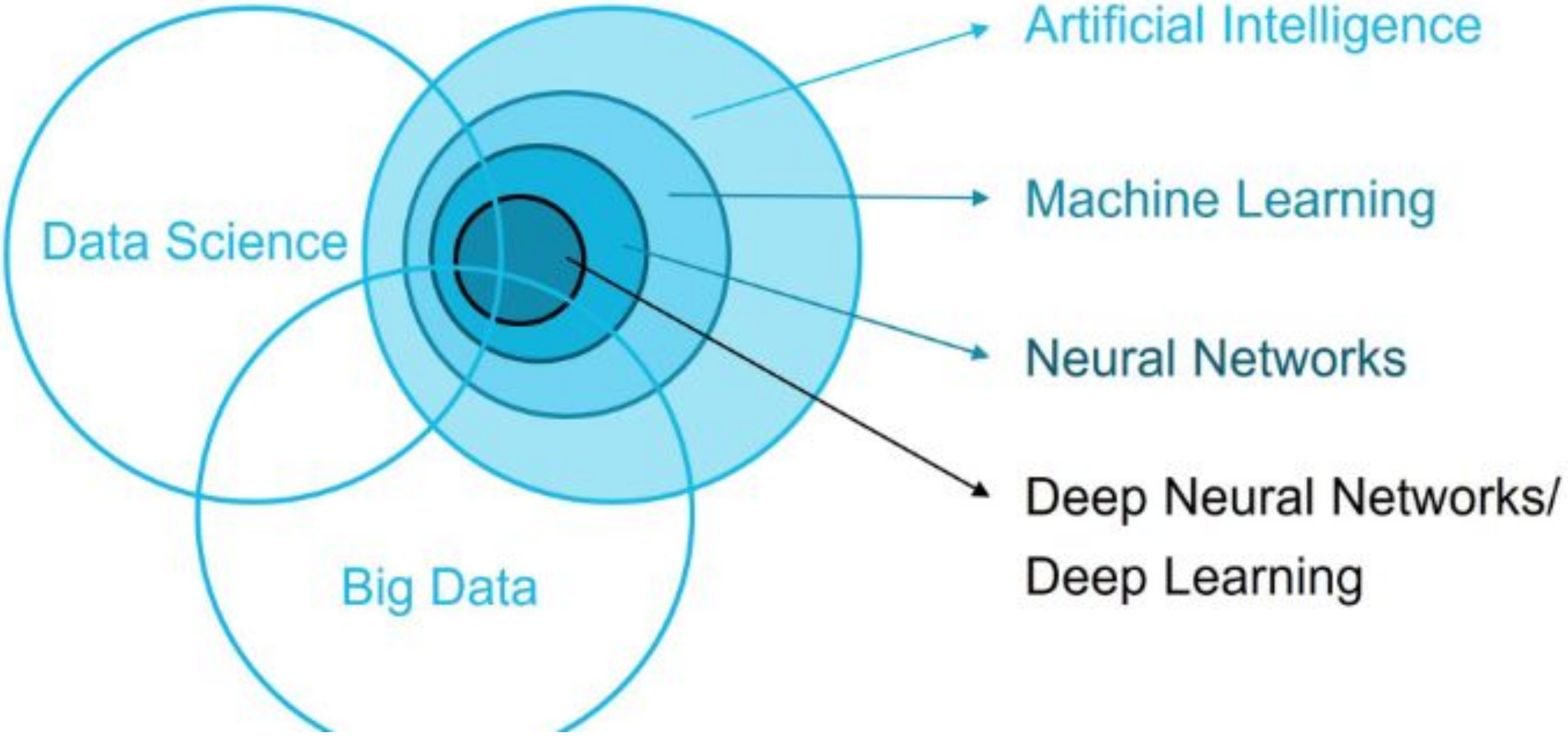
Text Book -Module 1

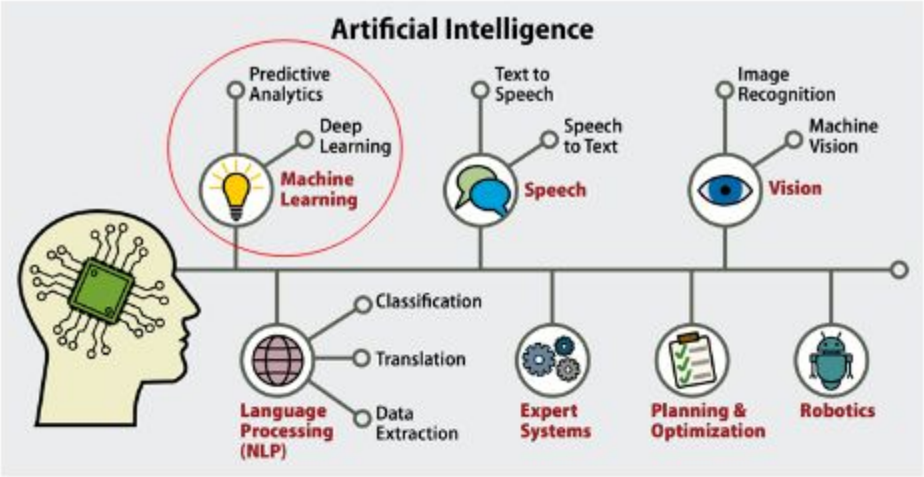


1.Introduction to Data Science and Machine Learning

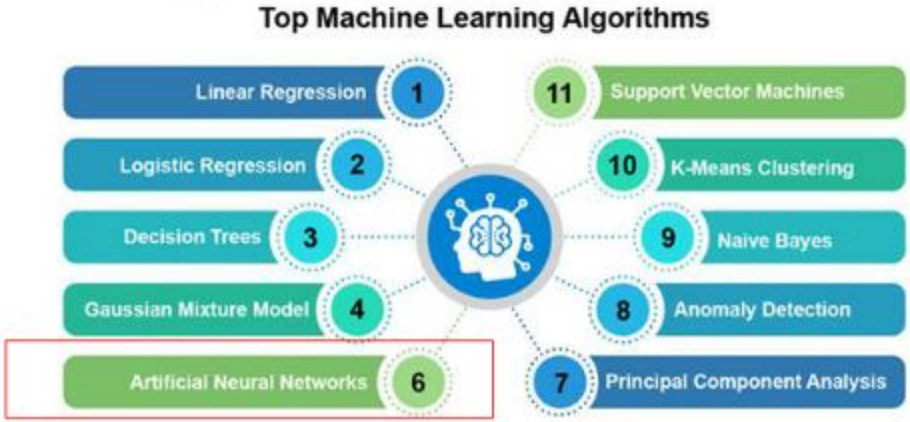


Confused ?

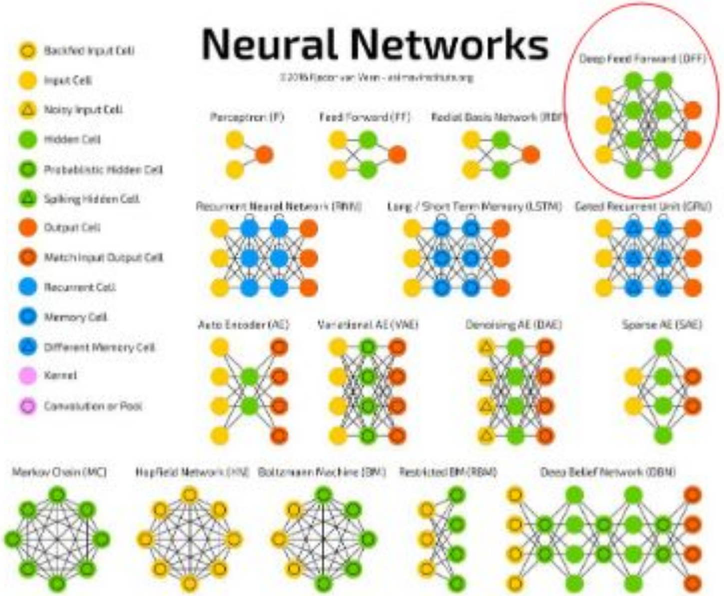

Data Science
Big Data
Artificial Intelligence
Machine Learning
Neural Networks
Deep Neural Networks/
Deep Learning



Artificial Intelligence
Predictive Analytics
Deep Learning
Machine Learning
Text to Speech
Speech to Text
Speech
Image Recognition
Machine Vision
Vision
Classification
Translation
Data Extraction
Language Processing (NLP)
Expert Systems
Planning & Optimization
Robotics

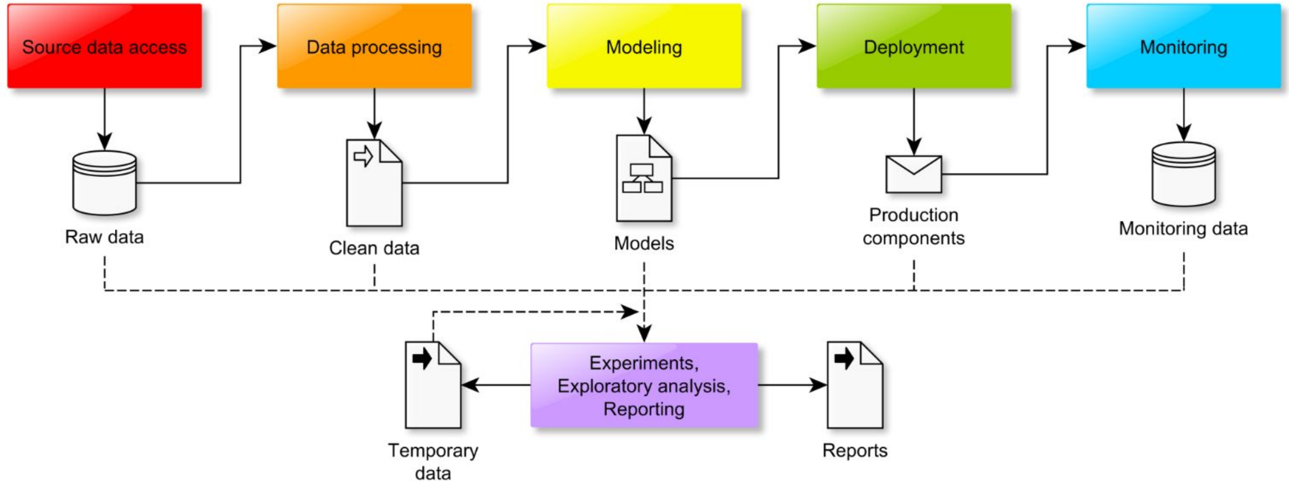


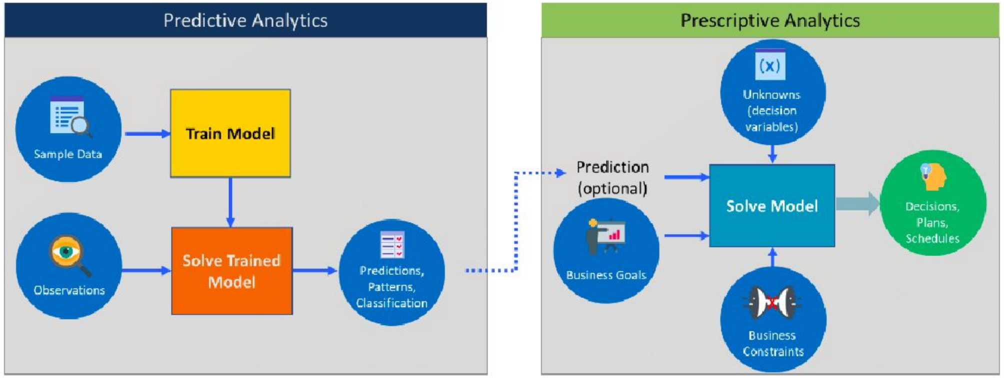
Top Machine Learning Algorithms
Linear Regression 1
Logistic Regression 2
Decision Trees 3
Gaussian Mixture Model 4
Artificial Neural Networks 6
11 Support Vector Machines
10 K-Means Clustering
9 Naive Bayes
8 Anomaly Detection
7 Principal Component Analysis



Neural Networks
Backfed Input Cell
Input Cell
Noisy Input Cell
Hidden Cell
Probablistic Hidden Cell
Spiking Hidden Cell
Output Cell
Match Input Output Cell
Recurrent Cell
Memory Cell
Different Memory Cell
Kernel
Convolution or Pool
Deep Feed Forward (DFF)
Perceptron (P)
Feed Forward (FF)
Recurrent Neural Network (RNN)
Long / Short Term Memory (LSTM)
Gated Recurrent Unit (GRU)
Auto Encoder (AE)
Variational AE (VAE)
Denoising AE (DAE)
Sparse AE (SAE)
Markov Chain (MC)
Boltzmann Machine (BM)
Restricted BM (RBM)
Deep Belief Network (DBN)

Data Science Pipeline





Programming vs Learning

Traditional Programming

Machine Learning

1.1 AI, MACHINE LEARNING, AND DATA SCIENCE

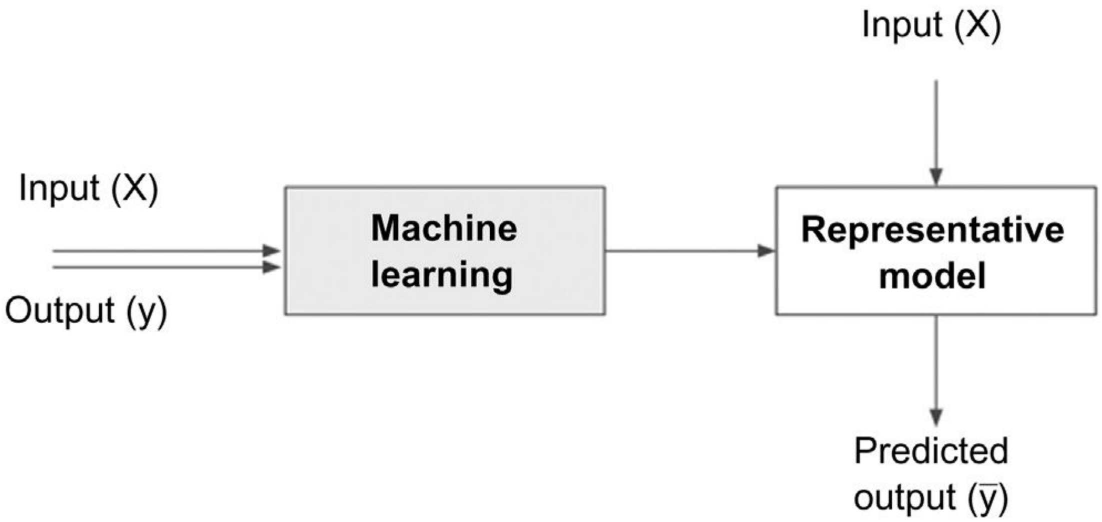
- Artificial intelligence is about giving machines the capability of mimicking human behavior, particularly cognitive functions.

- Examples would be: facial recognition, automated driving, sorting mail based on postal code.

- **Machine learning** can either be considered a sub-field or one of the tools of artificial intelligence, is providing machines with the capability of learning from experience. Experience for machines comes in the form of data.
- Data that is used to teach machines is called training data

Machine learning algorithms,also called **“learners”**, take both the known input and output (training data) to figure out a model for the program which converts input to output.

Data science models



Data Science-Definition

- **1.2 WHAT IS DATA SCIENCE?**

Data science is the business application of machine learning, artificial intelligence, and other quantitative fields like statistics, visualization, and mathematics. It is an interdisciplinary field that extracts value from data.Data science is the domain of study that deals with vast volumes of data using modern tools and techniques to find unseen patterns, derive meaningful information, and make business decisions

- Data science starts with data, which can range from a simple array of a few numeric observations to a complex matrix of millions of observations with thousands of variables. Data science utilizes certain specialized computational methods in order to discover meaningful and useful structures within a dataset.
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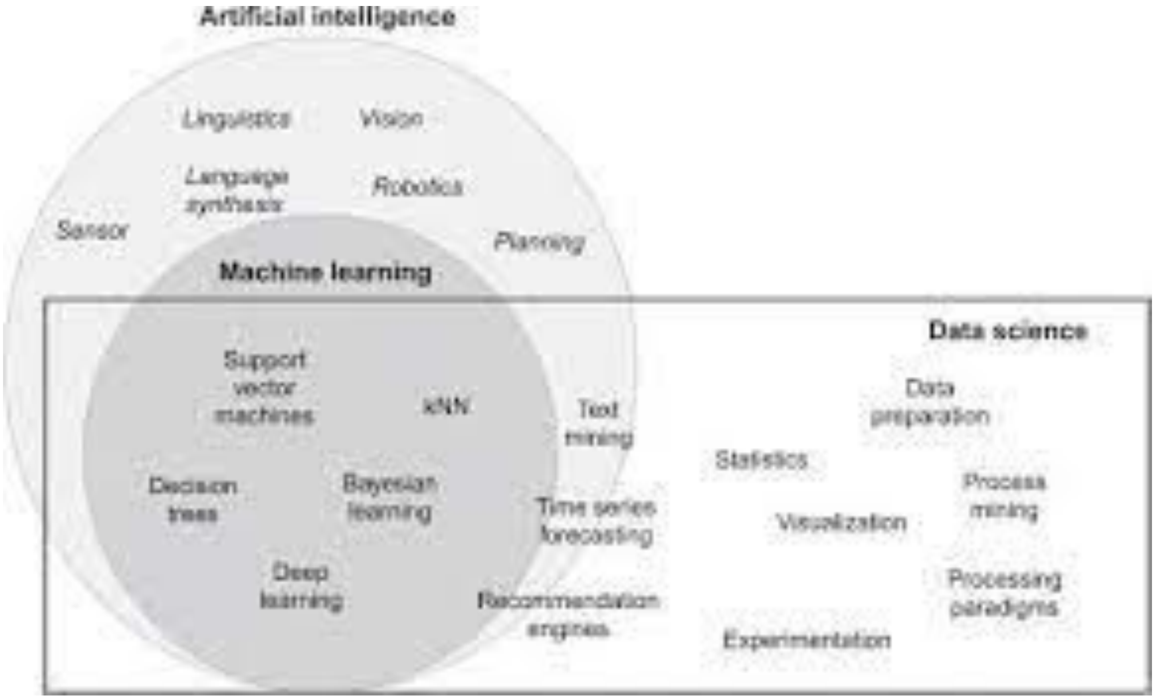
- The discipline of data science coexists and is closely associated with a number of related areas such as database systems, data engineering, visualization, data analysis, experimentation, and business intelligence (BI).

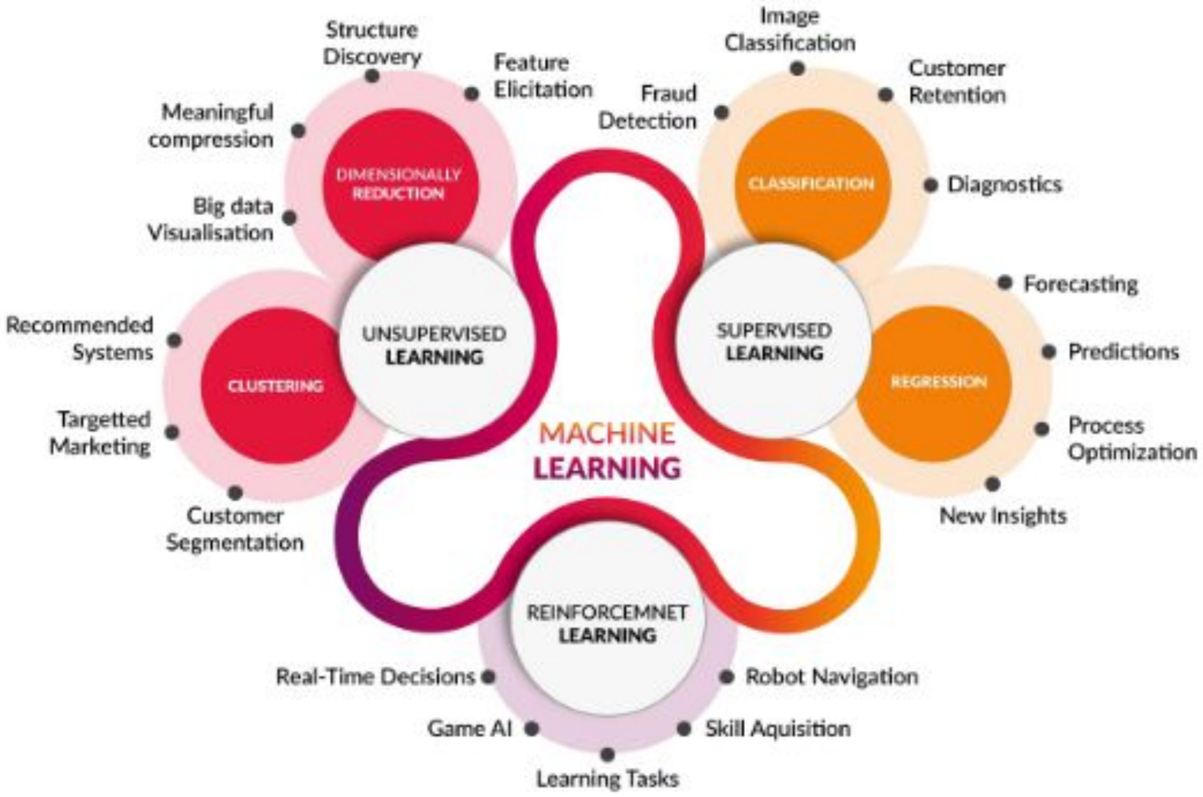
Artificial intelligence, machine learning, and data science.



1.3 DATA SCIENCE CLASSIFICATION

- Data science problems can be broadly categorized into
- **Supervised**
- **Unsupervised learning models.**
- Supervised or directed data science tries to infer a function or relationship based on labeled training data and uses this function to map new unlabeled data.
- Supervised techniques predict the value of the output variables based on a set of input variables. To do this, a model is developed from a **training dataset where the values of input and output are previously known.**
- The model generalizes the relationship between the input and output variables and uses it to predict for a dataset where only input variables are known. **The output variable that is being predicted is also called a class label or target variable**

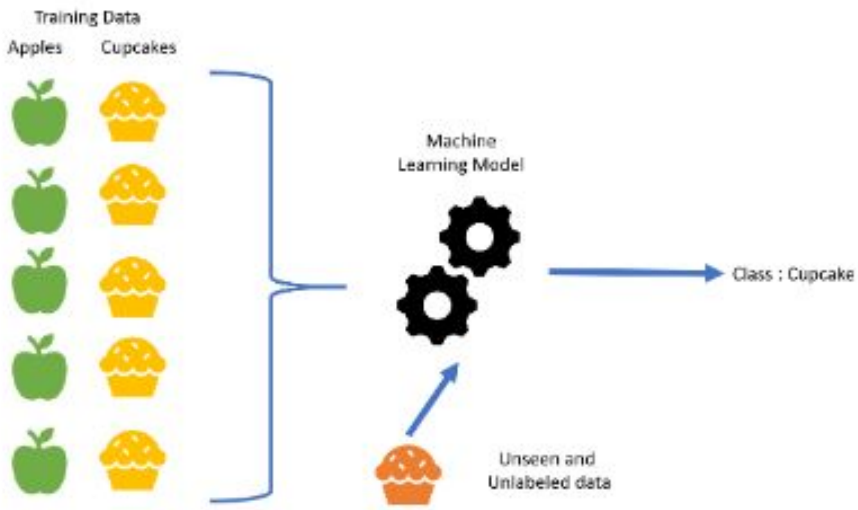
Types of Learning



Supervised Learning

Input : Labeled Data

X $(features)$	Y $(labels)$
$x_{11}, x_{12}, x_{13}, \dots \dots \dots \dots \dots \dots x_{1n}$	y_1
$\vdots$	$\vdots$
$x_{k1}, xk_2, x_{k3}, \dots \dots \dots \dots \dots \dots x_{kn}$	y_k



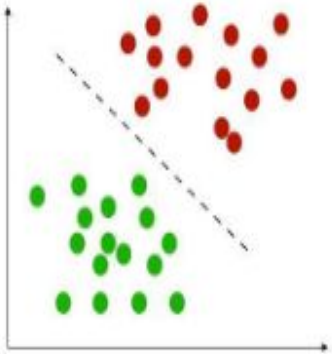
Goal : Construct a predictor $f : X \rightarrow Y$

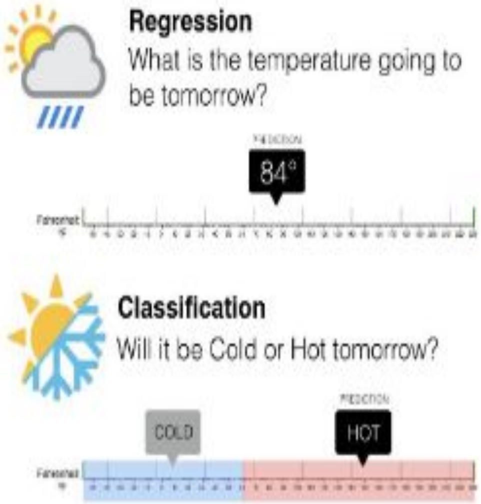
to minimize the error between $\widehat{Y}, Y$
where, $\widehat{Y} = f(X)$

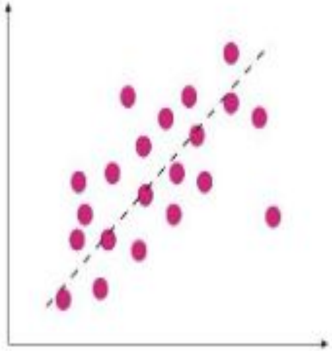
Use : using predictor to predict $\widehat{y} = f(\widehat{x})$

For the unknown input $\hat{x}$

Supervised Learning







Classification

- features and discrete labels
- maps an input to discrete label(class)
- Eg: spam or not, type of cancer

Regression

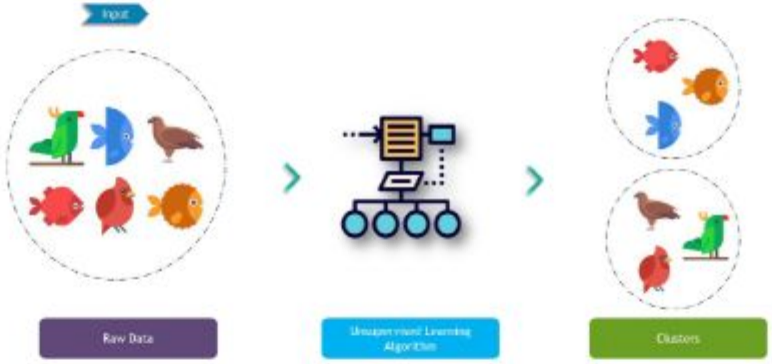
- features and continues real values
- Predict a real value for an input
- Eg: gold price, temperature

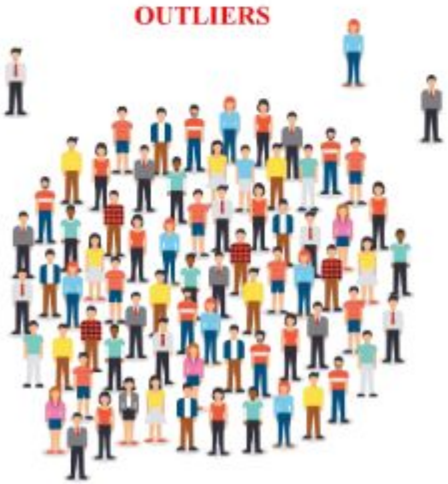
- Unsupervised or undirected data science **uncovers hidden patterns in unlabeled data.** In unsupervised data science, there are no output variables to predict.
- The objective of this class of data science techniques, is to find patterns in data based on the relationship between data points themselves.

Unsupervised Learning

Input : Unlabeled Data

X (*features*)
$x_{11}, x_{12}, x_{13}, \dots\dots\dots\dots\dots\dots\dots x_{1n}$
$\vdots$
$x_{k1}, xk_2, x_{k3}, \dots\dots\dots\dots\dots\dots\dots x_{kn}$



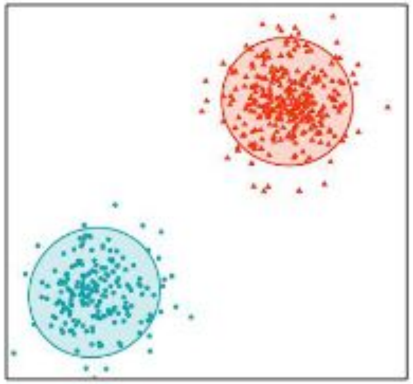


Goal : Construct a analyzer to find the hidden relationship between inputs

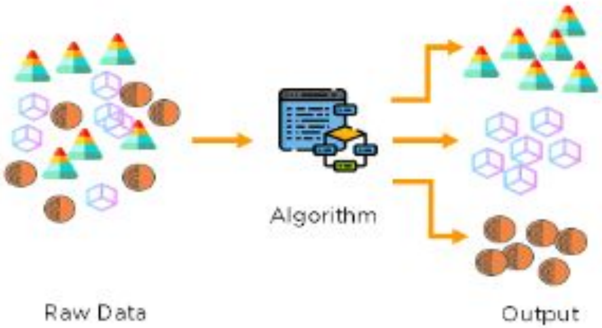
$$x_1, x_2, x_3, \dots, x_k$$

Use : Group or associate inputs according to their similarity

Unsupervised Learning



Clustering

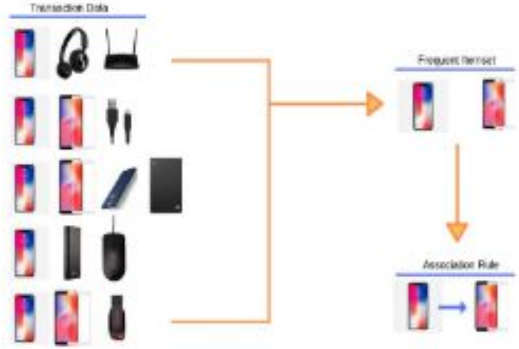


ID	Items
1	{Bread, Milk}
2	{Bread, Diapers, Beer, Eggs}
3	{Milk, Diapers, Beer, Cola}
4	{Bread, Milk, Diapers, Beer}
5	{Bread, Milk, Diapers, Cola}
...	...

market basket transactions

{Diapers, Beer} Example of a frequent itemset

{Diapers} → {Beer} Example of an association rule

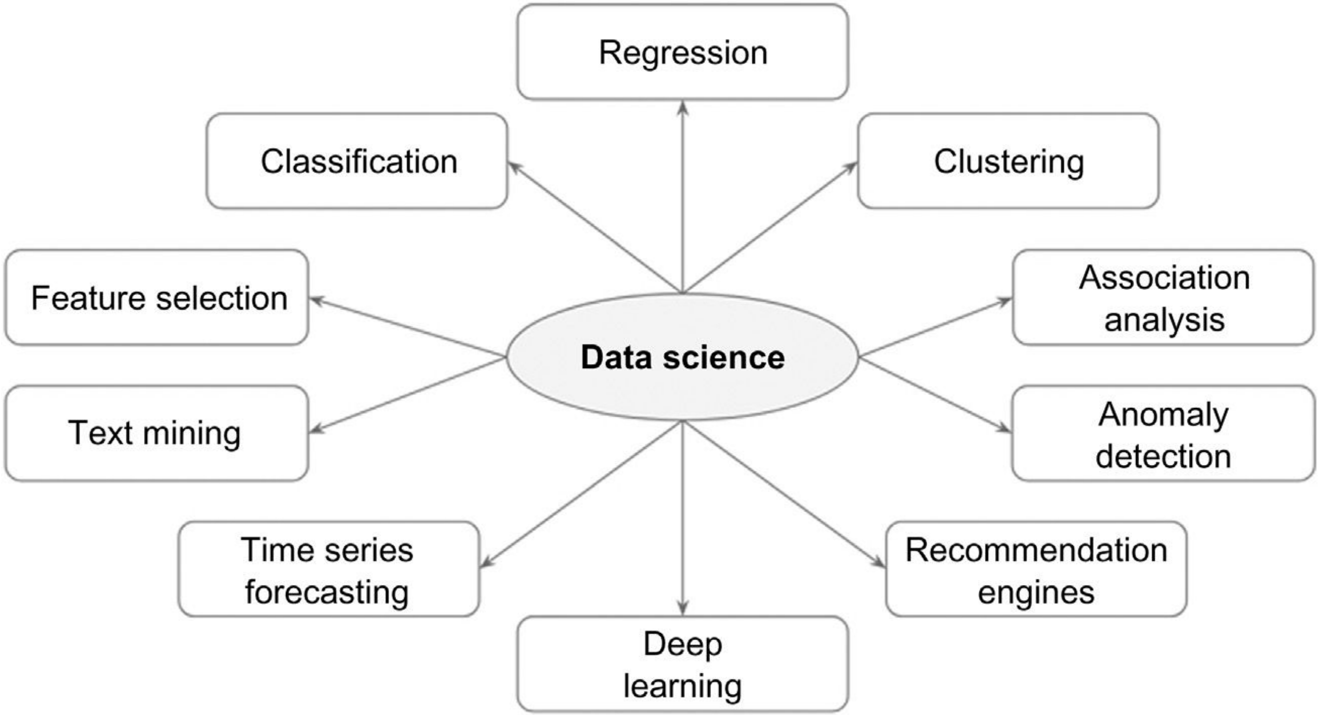


Itemset and rule mining

2.Data Science Process

- The methodical discovery of useful relationships and patterns in data is enabled by a set of iterative activities collectively known as the data science process. The standard data science process involves
- (1) Understanding the problem
- (2) Preparing the data samples
- (3) Developing the model
- (4) Applying the model on a dataset to see how the model may work in the real world
- (5) Deploying and maintaining the models.

Data Science Tasks



Data Science Frameworks

- A data science framework refers to a collection of tools and libraries designed to help a data science team discover, collect, organize, clean, and interpret data. Data science frameworks typically include tools and libraries for:

- Data collection and data acquisition
- Data visualization
- Statistical analysis
- Building machine learning models
- Numerical computing
- Data mining
- Building neural networks

Data Science Frameworks

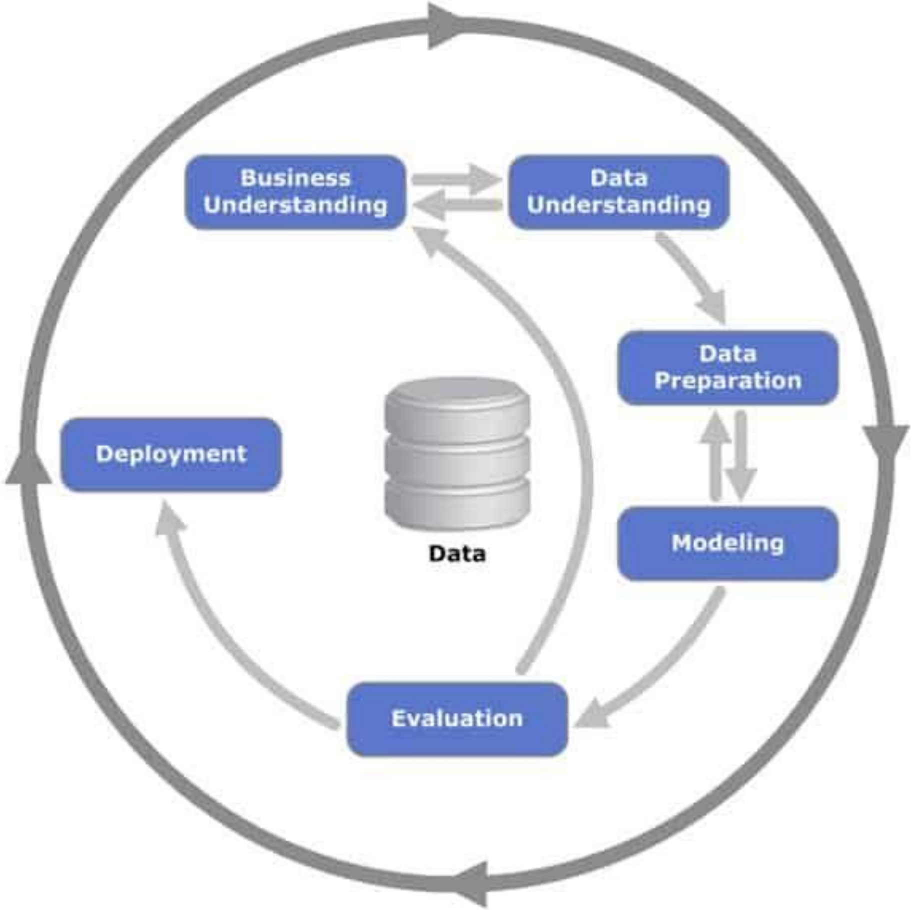
- Data science frameworks help data science teams
- **Save time**
- **Improve collaboration**
- **uncover patterns and insights.**

Data Science frameworks

- One of the most popular data science process frameworks is Cross Industry Standard Process for Data Mining
- (**CRISP-DM)**, which is an acronym for Cross Industry Standard Process for Data Mining. This framework was developed by a consortium of companies involved in data mining (Chapman et al., 2000).
- The CRISP-DM process is the most widely adopted framework for developing data science solution

- Other data science frameworks are **SEMMA,** an acronym for Sample, Explore, Modify, Model, and Assess, developed by the SAS Institute (SAS Institute, 2013);
- **DMAIC**, is an acronym for Define,Measure, Analyze, Improve, and Control, used in Six Sigma practice (Kubiak & Benbow, 2005); and the Selection, Preprocessing, Transformation, Data Mining, Interpretation, and Evaluation framework used in the knowledge discovery in databases process

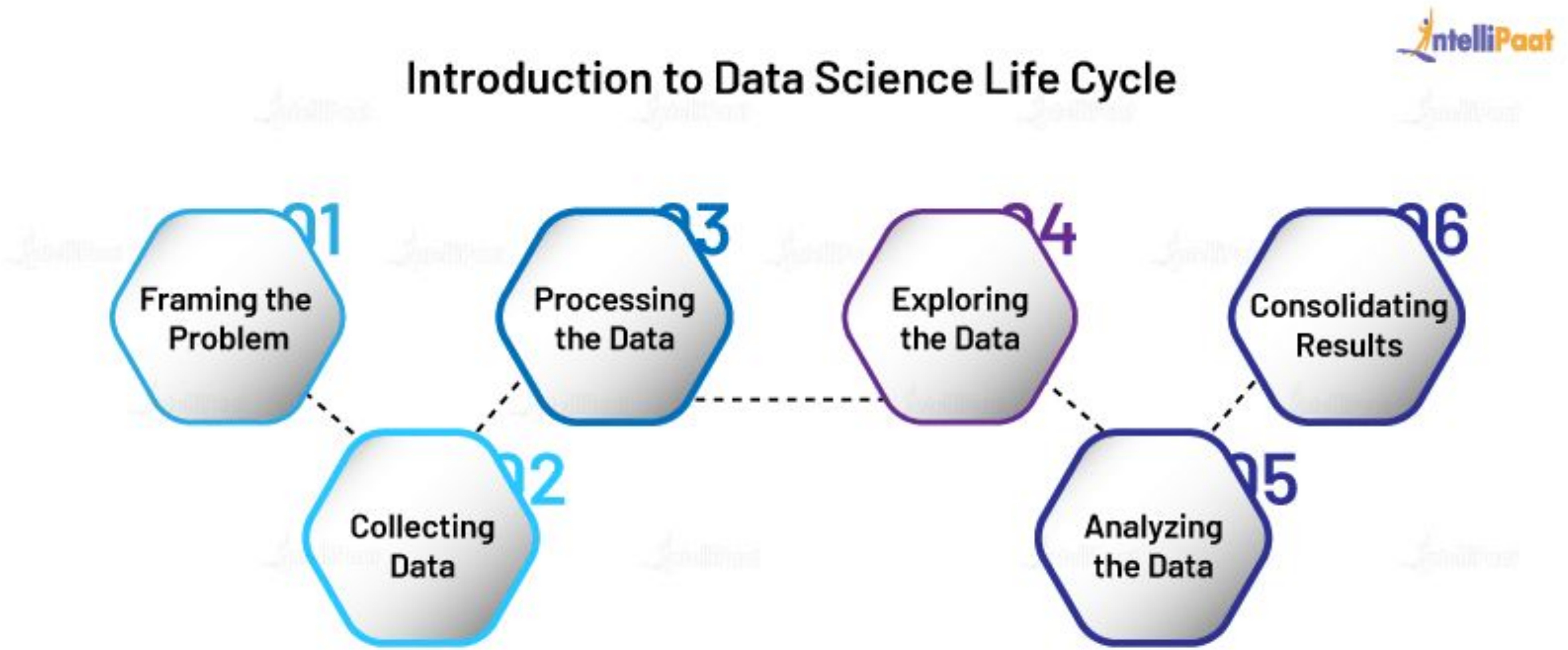
Crisp -Data mining framework

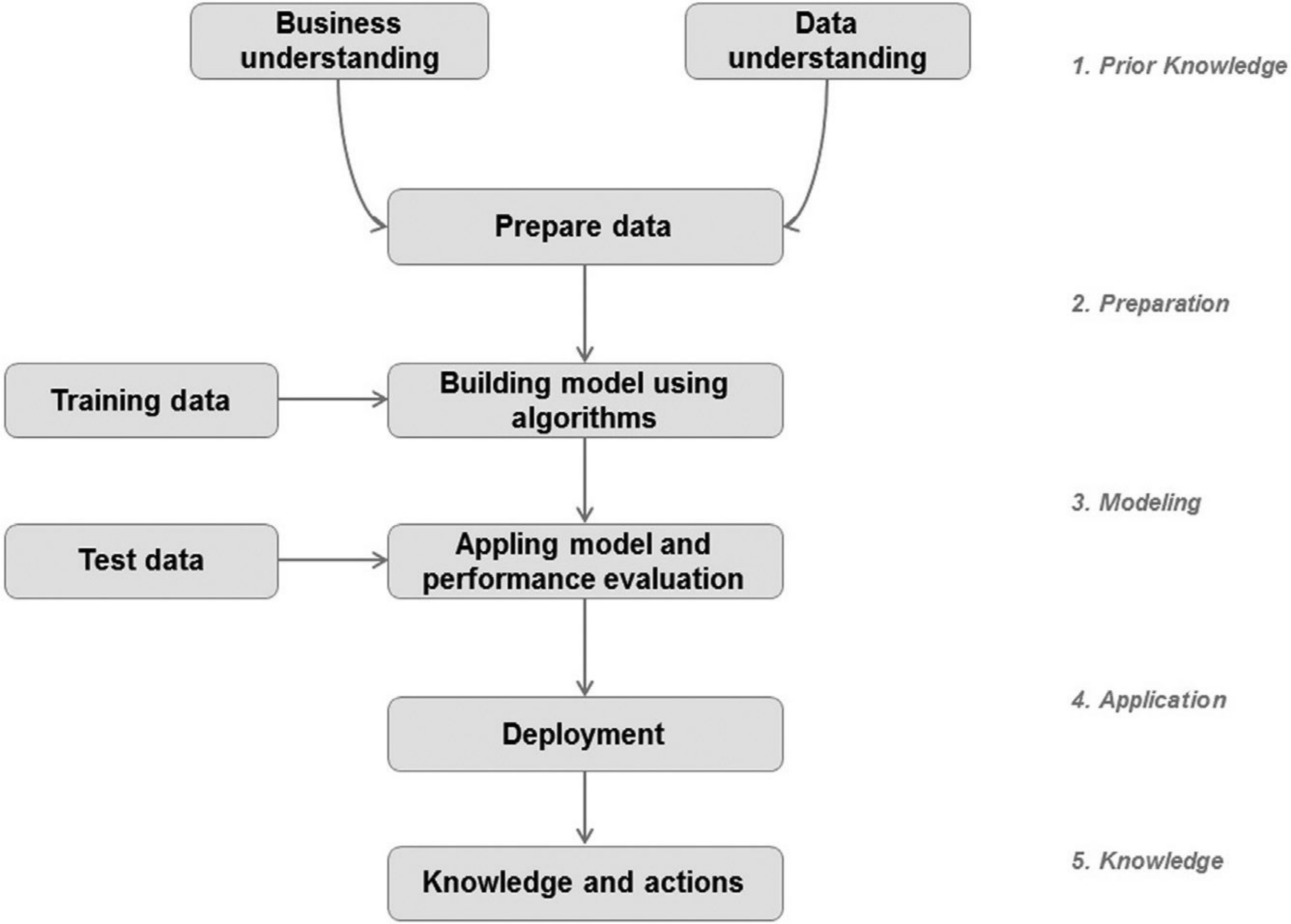


2.Data Science process

- The data science process presented in Fig. 2.2 is a generic set of steps that is problem, algorithm, and, data science tool agnostic.
- The fundamental **objective of any process that involves data science is to address the analysis question.** The problem at hand could be a segmentation of customers, a prediction of climate patterns, or a simple data exploration. The learning algorithm used to solve the business question could be a decision tree, an artificial neural network, or a scatter plot.
- The software tool to develop and implement the data science algorithm used could be custom coding, Rapid Miner, R, Weka, SAS,Oracle Data Miner, Python, etc.

Steps in Data Science process




Business understanding
Data understanding
1. Prior Knowledge
Prepare data
2. Preparation
Training data
Building model using algorithms
3. Modeling
Test data
Appling model and performance evaluation
4. Application
Deployment
5. Knowledge
Knowledge and actions

- **2.1.Prior Knowledge**
- Prior knowledge refers to information that is already known about a subject. The data science problem doesn’t emerge in isolation; it always develops on top of existing subject matter and contextual information that is already known.
- The prior knowledge step in the data science process helps to define what problem is being solved, how it fits in the business context, and what data is needed in order to solve the problem.

- **2.1.1 Objective**
- The data science process starts with a need for analysis, a question, or a business objective. This is possibly the most important step in the data science process .
- Without a well-defined statement of the problem, it is impossible to come up with the right dataset and pick the right data science algorithm.
- As an iterative process, it is common to go back to previous data science process steps, revise the assumptions, approach, and tactics

- **2.1.2.Subject Area**
- The process of data science uncovers hidden patterns in the dataset by exposing relationships between attributes. But the problem is that it uncovers a lot of patterns.
- **2.1.3.Data**
- Similar to the prior knowledge in the subject area, prior knowledge in the data can also be gathered. Understanding how the data is collected, stored,transformed, reported, and used is essential to the data science process.
- This part of the step surveys all the data available to answer the business question and narrows down the new data that need to be sourced. There are quite a range of factors to consider: **quality of the data, quantity of data, availability of data, gaps in data, does lack of data** compel the practitioner to change the business question, etc. **The objective of this step is to come up with a dataset to answer the business question through the data science process.**

Terminologies in Data science process

- A **dataset** is a collection of data with a defined structure.
- A **data point** (record, object or example) is a single instance in the dataset .Each row in Table is a data point
- An **attribute** (feature, input, dimension, variable, or predictor) is a single property of the dataset. Each column is an attribute.
- A **label** (class label, output, prediction, target, or response) is the special attribute to be predicted based on all the input attributes Identifiers are special attributes that are used for locating or providing context to individual records.
- For example, common attributes like names, account numbers, and employee ID numbers are identifier attributes

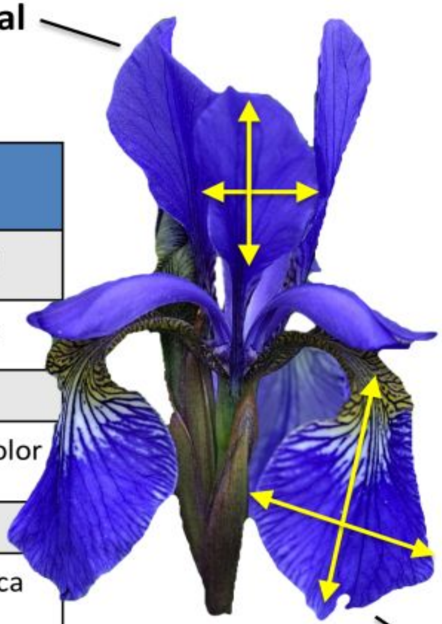
Iris Dataset

Samples
(instances, observations)

	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
			...		
50	6.4	3.5	4.5	1.2	Versicolor
			...		
150	5.9	3.0	5.0	1.8	Virginica

Features
(attributes, measurements, dimensions)

Class labels
(targets)



2.1.4 Causation v/s Correlation

- The correlation between the input and output attributes doesn’t guarantee causation. Hence, it is important to frame the data science question correctly using the existing domain and data knowledge.
- A c**orrelation between variables, however, does not automatically mean that the change in one variable is the cause of the change in the values of the other variable.** Causation indicates **that one event is the result of the occurrence of the other event**

- 2.2 DATA PREPARATION
- Preparing the dataset to suit a data science task is the most time-consuming part of the process. It is extremely rare that datasets are available in the form required by the data science algorithms.

2.2.1 Data Exploration

- Data preparation starts with an in-depth exploration of the data and gaining a better understanding of the dataset.
- Data exploration, also known as exploratory data analysis, provides a set of simple tools to achieve basic understanding of the data. Data exploration approaches involve computing **descriptive statistics and visualization of data**

2.2.2 Data quality

- Data quality is an ongoing concern wherever data is collected, processed, and stored.
- Organizations use data alerts, cleansing, and transformation techniques to improve and manage the quality of the data and store them in company wide repositories called data warehouses.

2.2.3 Missing values

- One of the most common data quality issues is that some records have missing attribute values
- The missing value can be substituted with a range of artificial data so that the issue can be managed with marginal impact on the later steps in the data science process.

2.2.4 Data types and conversion

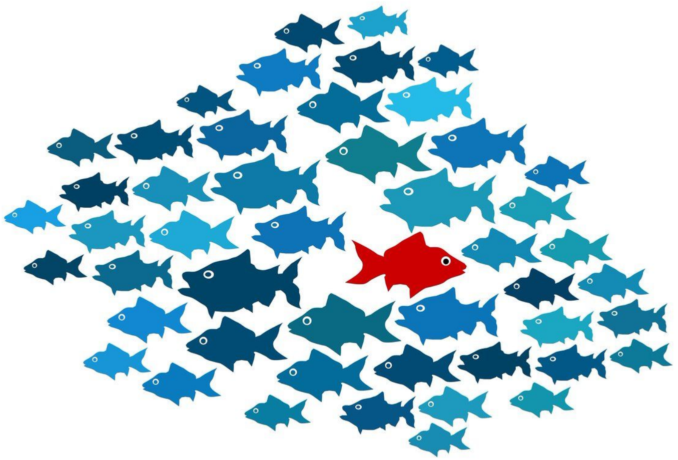
- The attributes in a dataset can be of different types, such as **continuous numeric (interest rate), integer numeric (credit score), or categorical.**
- If the available data are **categorical, they must be converted to continuous numeric attribute. A specific numeric score can be encoded for each category value, such as poor =400, good = 600, excellent = 700,** etc.
- Similarly, **numeric values can be converted to categorical data types by a technique called binning**, where a range of values are specified for each category, for example, a score between 400 and 500 can be encoded as “low” and so on.

2.2.5 Transformation

- In some data science algorithms like k-NN, the input attributes are expected to be **numeric and normalized,** because the algorithm compares the values of different attributes and calculates distance between the data points.

2.2.6 outliers

- Outliers are anomalies in a given dataset. Outliers may occur because of incorrect data capture.
- Regardless, the presence of outliers needs to be understood and will require special treatments



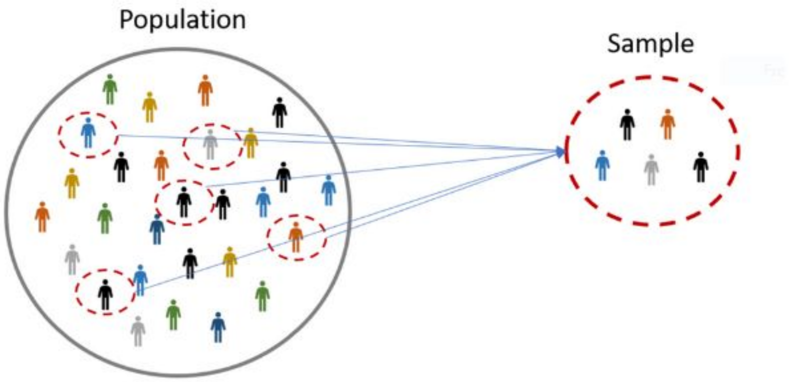
2.2.7 Feature selection

- A large number of attributes in the dataset significantly increases the complexity of a model and may degrade the performance of the model due to the curse of dimensionality.
- Reducing the number of attributes, without significant loss in the performance of the model, is called feature selection.

2.2.8 Data Sampling

- Sampling is a process of selecting a subset of records as a representation of the original dataset for use in data analysis or modeling.

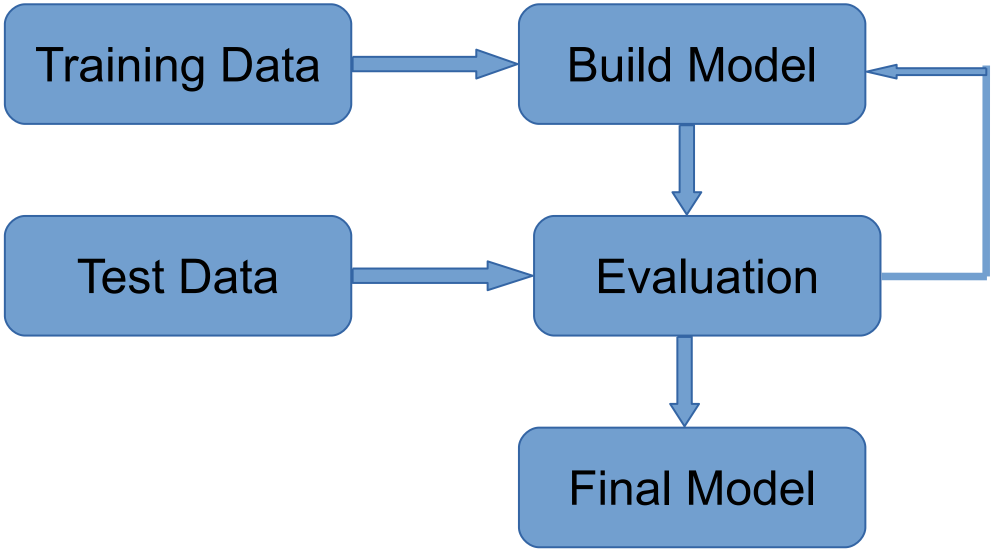
- The sample data serve as a representative of the original dataset with similar properties, such as a similar mean. Sampling reduces the amount of data that need to be processed and speeds up the build process of the modeling.



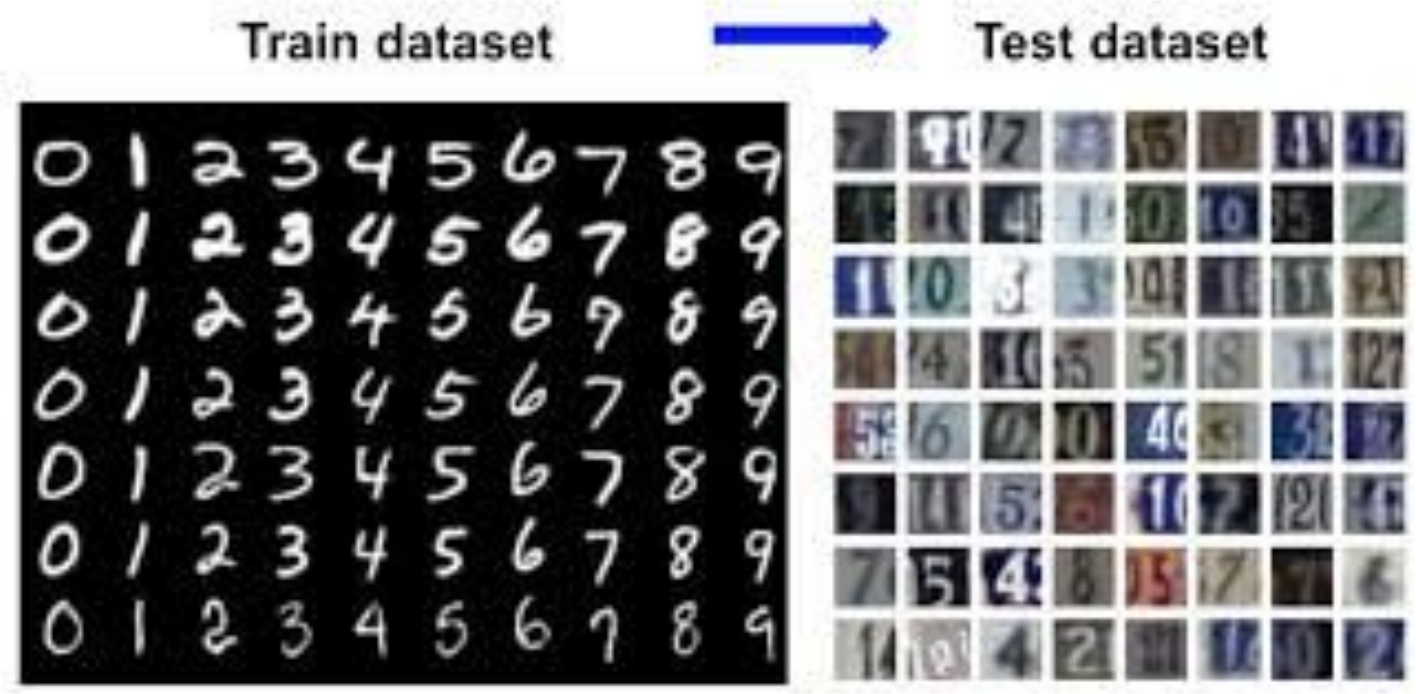
- **2.3 Modelling**

- A model is the abstract representation of the data and the relationships in a given dataset.
- **Classification and regression** tasks are predictive techniques because they **predict an outcome variable based on one or more input variables**. Predictive algorithms require a prior known dataset to learn the model.

Modelling steps



2.3.1 Train and Test datasets



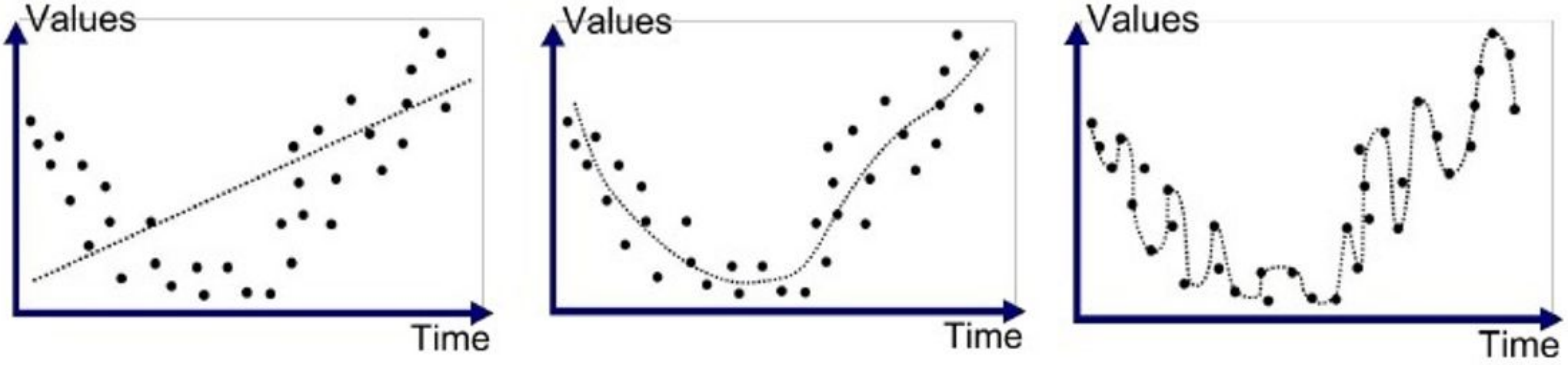
- The modelling step creates a representative model inferred from the data. The dataset used to create the model, with known attributes and target, is called the training dataset.
- The validity of the created model will also need to be checked with another known dataset called the test dataset or validation dataset.

2.3.2 Learning Algorithms

- Classification task many algorithms can be chosen from: decision trees, rule induction, neural networks, Bayesian models, k-NN, etc.
- Likewise, within decision tree techniques, there are quite a number of variations of learning algorithms like classification and regression tree (CART), CHi-squared Automatic Interaction Detector (CHAID) etc

2.3.3 Model Evaluation

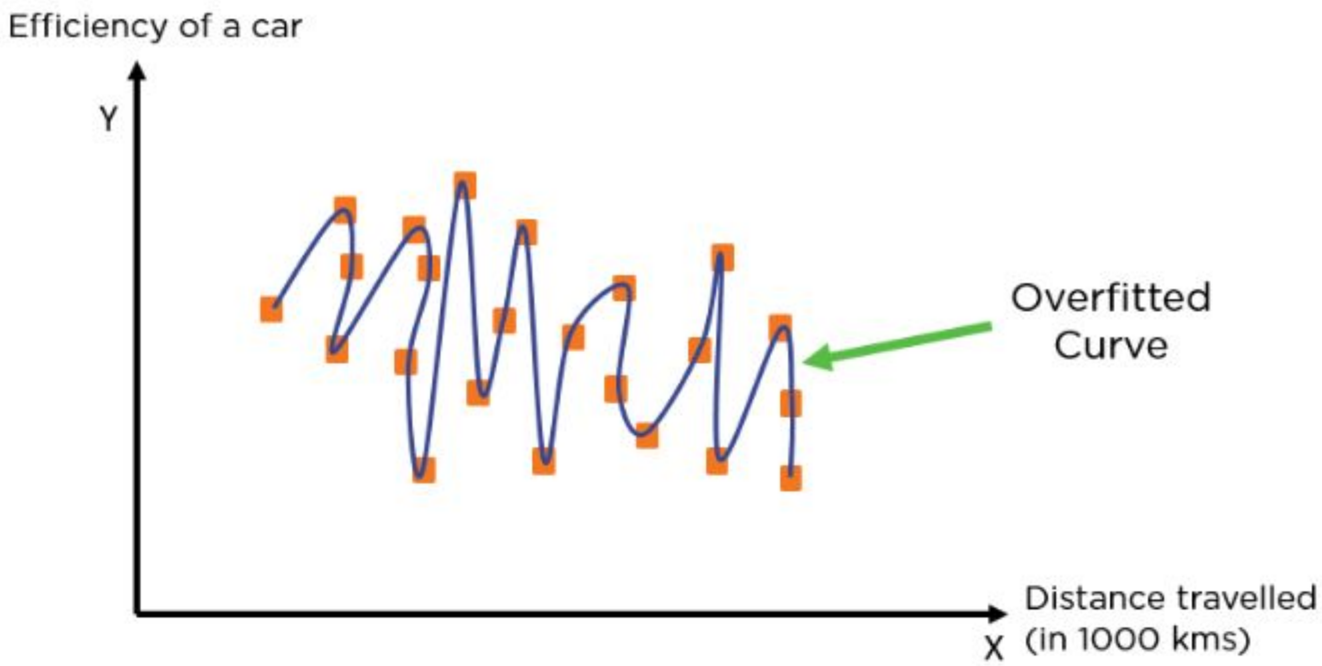
- The estimation may not be exactly the same as the values in the training records. A model should not memorize and output the same values that are in the training records.
- The phenomenon of a model memorizing the training data is called overfitting. An overfitted model just memorizes the training records and will underperform on real unlabeled new data.



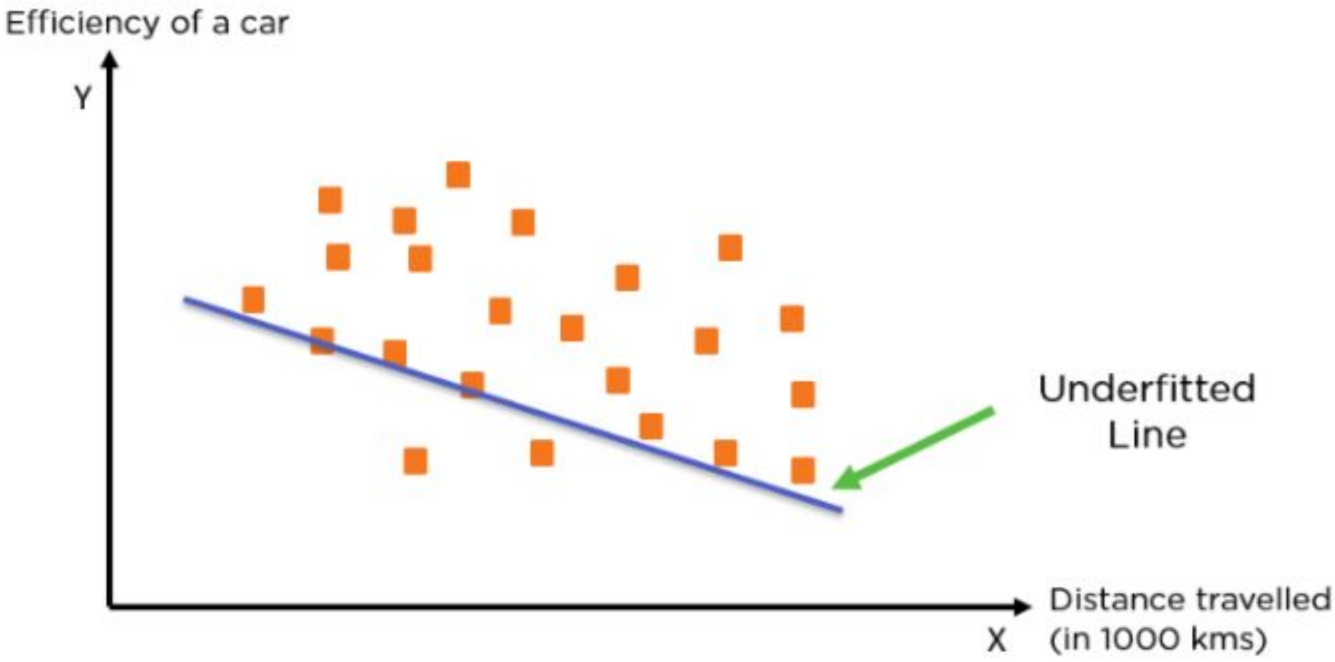
Underfitted　Good Fit/Robust　Overfitted

Overfitting

- When a model performs very well for training data but has poor performance with test data (new data), it is known as overfitting.
- In this case, the machine learning model learns the details and noise in the training data such that it negatively affects the performance of the model on test data.


Efficiency of a car
Y
Overfitted
Curve
Distance travelled
(in 1000 kms)
X

- What is Underfitting?
- When a model has not learned the patterns in the training data well and is unable to generalize well on the new data, it is known as underfitting. An underfit model has poor performance on the training data and will result in unreliable predictions.


Efficiency of a car
Y
Underfitted
Line
X
Distance travelled
(in 1000 kms)

2.3.4 Ensemble Modelling

- Ensemble modeling is a process where multiple diverse base models are used to predict an outcome.Ensemble methods is a machine learning technique that combines several base models in order to produce one optimal predictive model.
- The motivation for using ensemble models is to reduce the generalization error of the prediction. As long as the base models are diverse and independent, the prediction error decreases when the ensemble approach is used

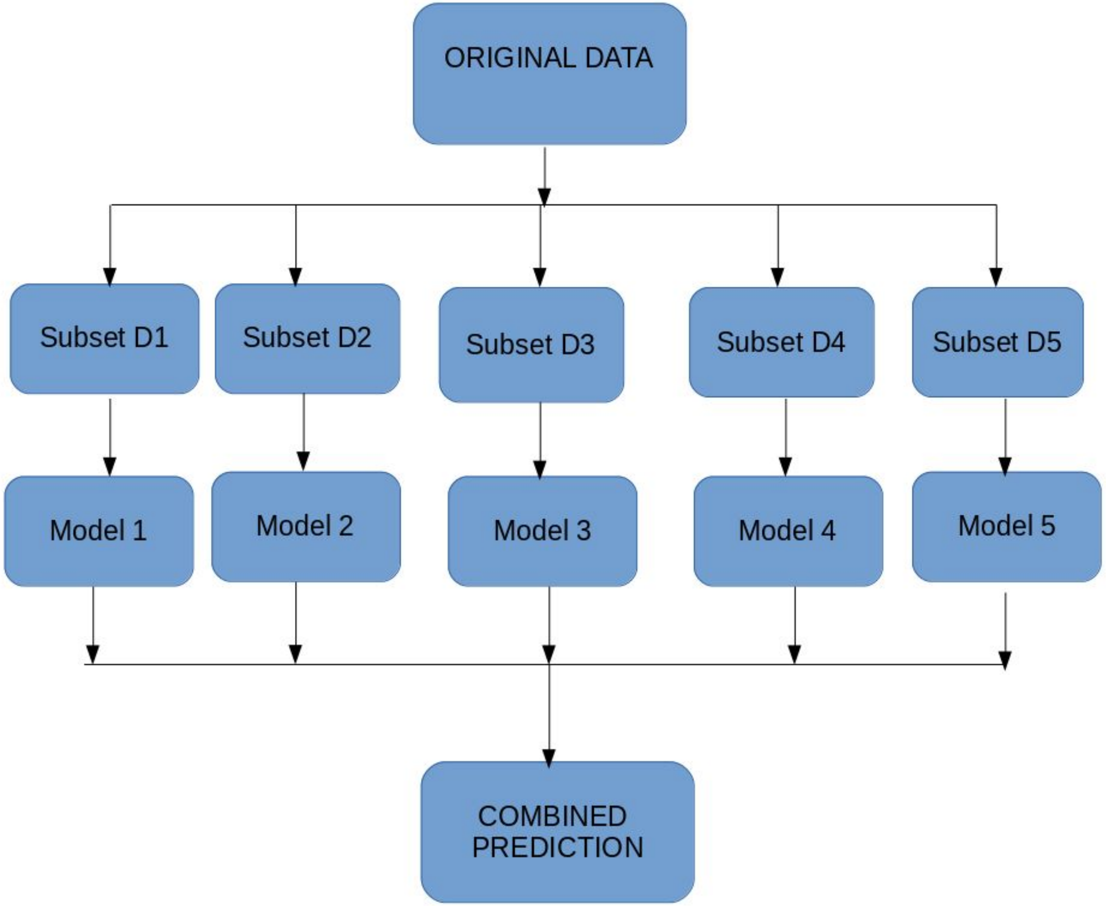
Ensemble modelling

Ensemble models in machine learning operate on a similar idea. They **combine the decisions from multiple models to improve the overall performance.**

Ensemble modelling methods

- 2.1 Max Voting
- The max voting method is generally used for classification problems. In this technique, multiple models are used to make predictions for each data point. The predictions by each model are considered as a ‘vote’. The predictions which we get from the majority of the models are used as the final prediction.

Ensemble modelling



- The result of max voting would be something like this:
- **Colleague 1 Colleague 2 Colleague 3 Colleague 4 Colleague 5 Final rating**
  - **5 4 5 4 4 4**

Averaging

- we take an average of predictions from all the models and use it to make the final prediction.
- Based on the previous example we get (5+4+5+4+4)/5 = 4.4
- Colleague 1 Colleague 2 Colleague 3 Colleague 4 Colleague 5 Final rating
  - 5 4 5 4 4 4.4

Weighted average

- This is an extension of the averaging method. All models are assigned different weights defining the importance of each model for prediction.
- For instance, if two of your colleagues are critics, while others have no prior experience in this field, then the answers by these two friends are given more importance as compared to the other people.

- The result is calculated as
- [(5*0.23) + (4*0.23) + (5*0.18) + (4*0.18) + (4*0.18)] = 4.41.
- Colleague 1 Colleague 2 Colleague 3 Colleague 4 Colleague 5 Final rating
- 0.23 0.23 0.18 0.18 0.18
- 5 4 5 4 4
- Result =4.41

Data science process summary

- At the end of the modeling stage of the data science process, one has
- (1) Analyzed the business question
- (2) Sourced the data relevant to answer the question
- (3) Selected a data science technique to answer the question
- (4) Picked a data science algorithm and prepared the data to suit the algorithm;
- (5) Split the data into training and test datasets
- (6) Built a generalized model from the training dataset
- (7) Validated the model against the test

2.4 Application

- Deployment is the stage at which the model becomes production ready or live
- 2.4.1 Production Readiness
- The production readiness part of the deployment determines the critical qualities required for the deployment objective.

- Consider two business use cases: determining whether a consumer qualifies for a loan and determining the groupings of customers for an enterprise by marketing function.
- In the first use case of determining whether a consumer qualifies for a loan,it is optimal to **provide a quick decision while also proving accurate.**
- The decision-making model has to collect data from the customer, integrate third-party data like credit history, and make a decision on the loan approval and terms in a matter of seconds. The critical quality of this model deployment is **real-time prediction.**

- **Segmenting customers based on their relationship with the company** is a thoughtful process where signals from various customer interactions are collected.
- The critical quality in this application is the ability to find unique patterns amongst customers, not the response time of the model.

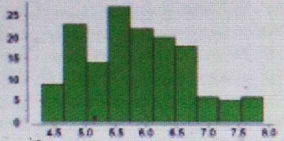
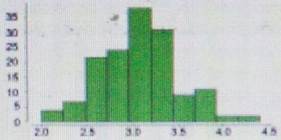
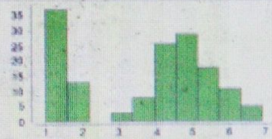
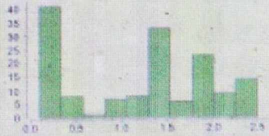
- 2.4.2 Technical Integration
- Currently, it is quite common to use data science automation tools or coding using R or Python to develop models.
- 2.4.3 Response Time
- Data science algorithms, like k-NN, are easy to build, but quite slow at pre-
- dicting the unlabeled records. Algorithms such as the decision tree take time
- to build but are fast at prediction. There are trade-offs to be made between
- production responsiveness and modeling build time.

- 2.4.4 Model Refresh
- The validity of the model can be routinely tested by using the new known test dataset and calculating the prediction error rate. If the error rate exceeds a particular threshold, then the model has to be refreshed and redeployed.

- 2.4.5 Assimilation
- In the descriptive data science applications, deploying a model to live systems may not be the end objective. The objective may be to assimilate the knowledge gained from the data science analysis to the organization.

3.Data Exploration

- Data preparation starts with an in-depth exploration of the data and gaining a better understanding of the dataset.
- Data exploration, also known as exploratory data analysis, provides a set of simple tools to achieve basic understanding of the data. Data exploration approaches involve computing **descriptive statistics and visualization of data.**

Sepal Length	Real	0	 Open chart	Min 4.300	Max 7.900	Average 5.843	Deviation 0.828
Sepal Width	Real	0	 Open chart	Min 2	Max 4.400	Average 3.054	Deviation 0.434
Petal Length	Real	0	 Open chart	Min 1	Max 6.900	Average 3.759	Deviation 1.764
Petal Width	Real	0	 Open chart	Min 0.100	Max 2.500	Average 1.199	Deviation 0.763

3.1 Objectives of Data Exploration

- **1.Data understanding**: Data exploration provides a high-level overview of each attribute (also called variable) in the dataset and the interaction between the attributes.

- **2.Data preparation:** Before applying the data science algorithm, the dataset has to be prepared for handling any of the anomalies that may be present in the data. These anomalies include outliers, missing values, or highly correlated attributes.

- **Data science tasks**: Basic data exploration can sometimes substitute the entire data science process. For example, scatter plots can **identify clusters** in low-dimensional data or **can help develop regression or classification models with simple visual rules**

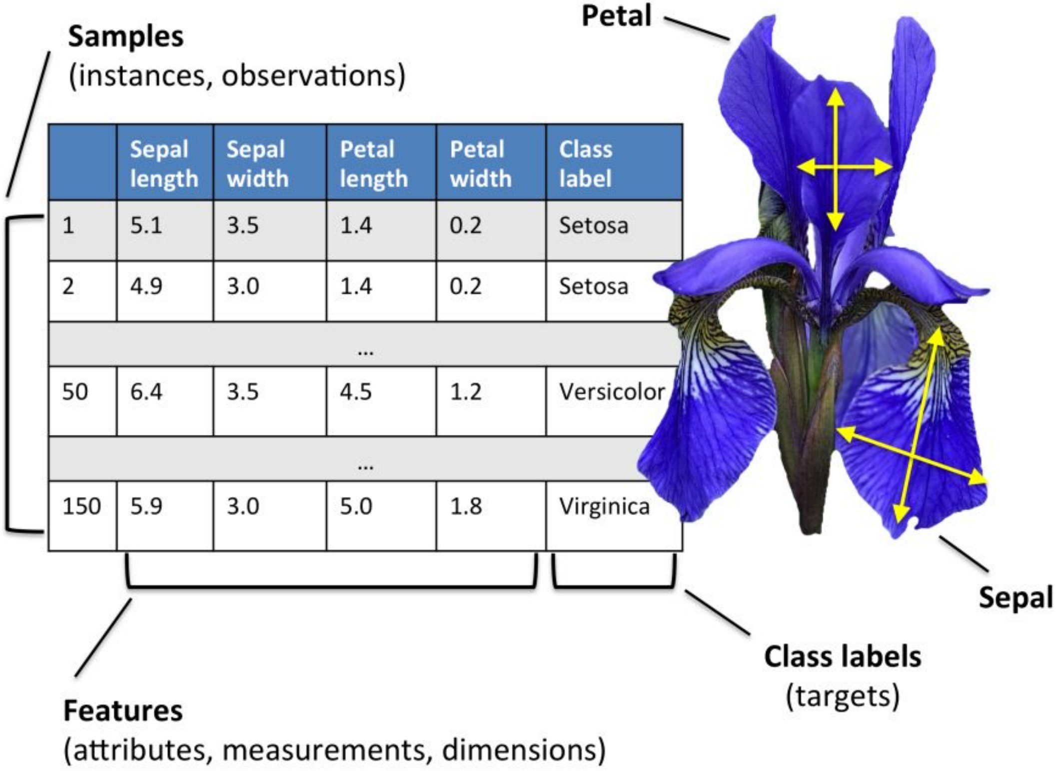
- **Interpreting the results**: Finally, data exploration is used in understanding the **prediction, classification, and clustering of the results of the data science process.**

3.2 Datasets

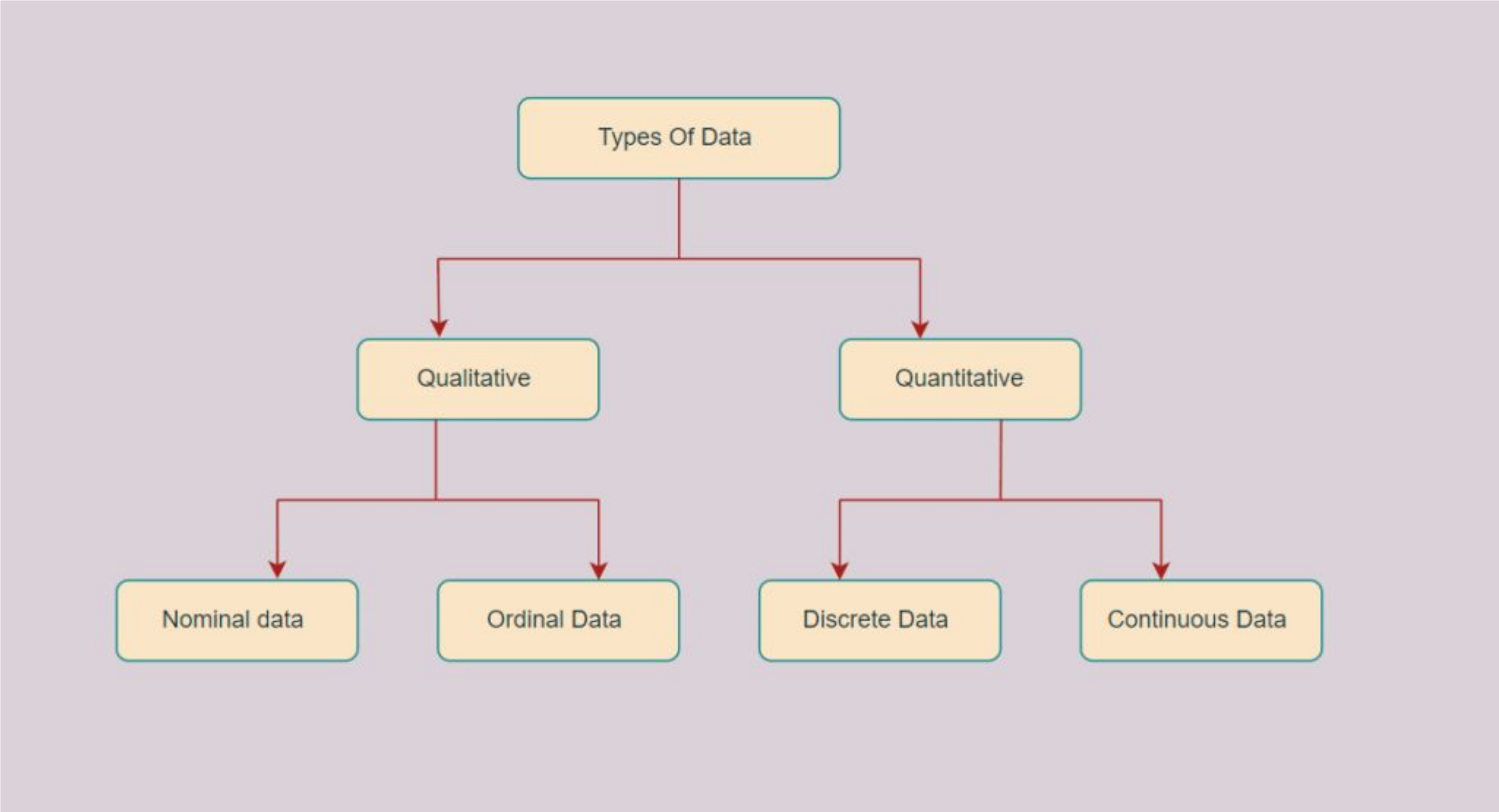
- The most popular datasets used to learn data science is probably the **Iris dataset**, introduced by Ronald Fisher.
- Iris is a flowering plant that is found widely, across the world. The genus of Iris contains more than 300 different species. Each species exhibits different physical characteristics like shape and size of the flowers and leaves.

- Each observation consists of four attributes: sepal length, sepal width, petal length, and petal width.

-



Types of Data



- **Numeric or Continuous**
- Temperature expressed in Centigrade or Fahrenheit is **numeric and continuous** because it can be denoted by numbers and take an infinite number of values between digits. Values are ordered and calculating the difference between the values makes sense.

- An **integer** is a special form of the numeric data type which does **not have decimals in the value** or more precisely does not have infinite values between consecutive numbers.
- **Categorical or Nominal**
- Categorical data types are attributes treate**d as distinct symbols or just names.** The color of the iris of the human eye is a categorical data type because it takes a value like black, green, blue, gray, etc.

- An ordered nominal (**Ordinal**) data type is a special case of a categorical data type where there is some kind of order among the values.
- An example of an ordered data type is temperature expressed as hot, mild, cold.

3.3 Descriptive statistics

- Descriptive statistics refers to the study of the aggregate quantities of a dataset. These measures are some of the commonly used notations in everyday life.

- Some examples of descriptive statistics include average annual income, median home price in a neighbourhood, range of credit scores of a population, etc. In general, descriptive analysis covers the following characteristics of the sample or population dataset.

Characteristics of the Dataset	Measurement Technique
Center of the dataset	Mean, median, and mode
Spread of the dataset	Range, variance, and standard deviation
Shape of the distribution of the dataset	Symmetry, skewness, and kurtosis

- **3.3.1 Univariate Exploration**
- Univariate data exploration denotes analysis of one attribute at a time. The example Iris dataset for one species, I. setosa, has 50 observations and 4 attributes.

Measure of Central Tendency

The objective of finding the central location of an attribute is to quantify the dataset with one central or most common number.

- **Mean**: The mean is the arithmetic average of all observations in the dataset. It is calculated by summing all the data points and dividing by the number of data points. The mean for sepal length in centimetres is 5.0060.
- **Median**: The median is the value of the central point in the distribution.The median is calculated by sorting all the observations from small to large and selecting the mid-point observation in the sorted list. If the number of data points is even, then the average of the middle two data points is used as the median. The median for sepal length is in centimeters is 5.0000.
- **Mode:** The mode is the most frequently occurring observation. In the dataset, data points may be repetitive, and the most repetitive data point is the mode of the dataset. In this example, the mode in centimeters is 5.1000.

Descriptive statistics for Iris Dataset

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

Descriptive statistics for Iris dataset

Table 3.1 Iris Dataset and Descriptive Statistics (Fisher, 1936)

Observation	Sepal Length	Sepal Width	Petal Length	Petal Width
1	5.1	3.5	1.4	0.2
2	4.9	3.1	1.5	0.1
...	...	...	...	...
49	5	3.4	1.5	0.2
50	4.4	2.9	1.4	0.2
Statistics	**Sepal Length**	**Sepal Width**	**Petal Length**	**Petal Width**
Mean	5.006	3.418	1.464	0.244
Median	5.000	3.400	1.500	0.200
Mode	5.100	3.400	1.500	0.200
Range	1.500	2.100	0.900	0.500
Standard deviation	0.352	0.381	0.174	0.107
Variance	0.124	0.145	0.030	0.011

- **Measure of Spread**
- In desert regions, it is common for the temperature to cross above 110 _x005F_x0001_ F during the day and drop below 30 _x005F_x0001_ F during the night while the average temperature for a 24-hour period is around 70 _x005F_x0001_ F.
- What matters here is not just the central location of the temperature, but the spread of the temperature. There are two common metrics to quantify spread.

- **Range:** The range is the difference between the maximum value and the minimum value of the attribute.
- In the example, the range for the temperature in the desert is 80 _x005F_x0001_ F and the range for the tropics is 20 _x005F_x0001_ F.
- The desert region experiences larger temperature swings as indicated by the range.

- **Deviation**: The variance and standard deviation measures the spread, by considering all the values of the attribute.
- Deviation is simply measured as the difference between any given value (x i) and the mean of the sample (μ).
- $$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$

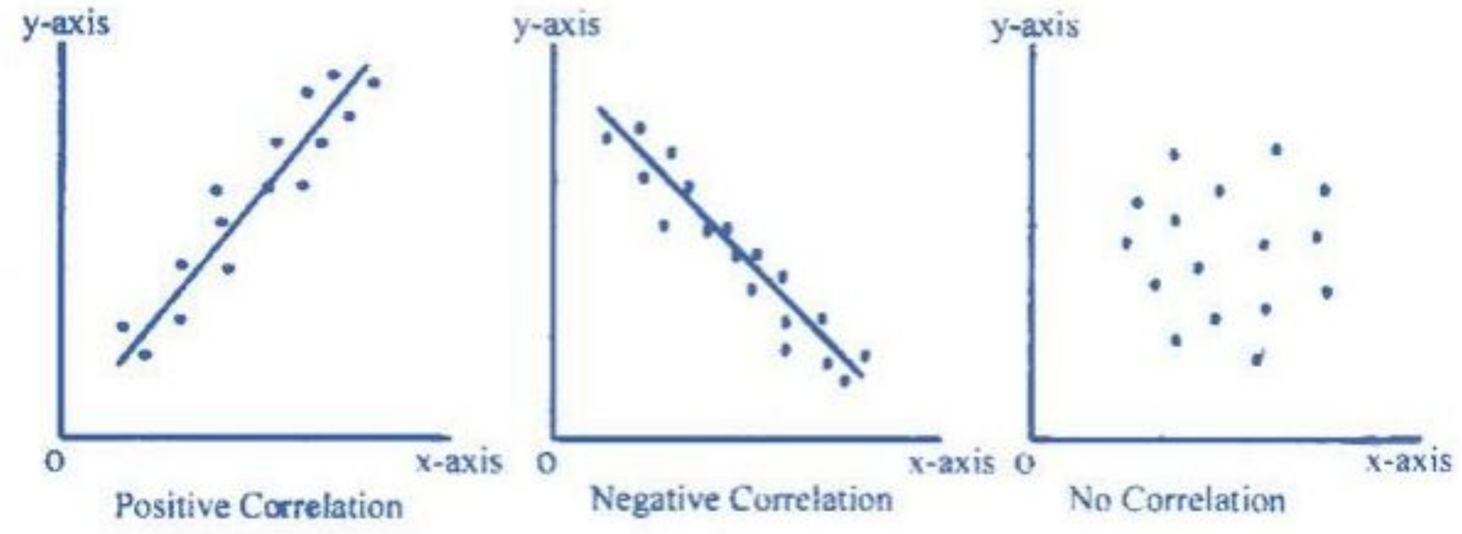
- The variance is the sum of the squared deviations of all data points divided by the number of data points. For a dataset with N observations, the variance is given by the following equation:
- $$\sigma^2 = \frac{\Sigma(xi - \bar{x})^2}{N}$$

- **3.3.2 Multivariate Exploration**
- Multivariate exploration is the study of m**ore than one attribute in the dataset simultaneously.** This technique is critical to understanding the relationship between the attributes
- **Central Data Point**
- In the Iris dataset, each data point as a set of all the four attributes can be expressed:
- observation i: {sepal length, sepal width, petal length, petal width}
- For example, observation one: {5.1, 3.5, 1.4, 0.2}. This observation point can also be expressed in four-dimensional Cartesian coordinates and can be plotted in a graph

- In this way, all 150 observations can be expressed in Cartesian coordinates. If the objective is to find the most “typical” observation point, it would be a data point made up of the mean of each attribute in the dataset Independently.
- For the Iris dataset the central mean point is {5.006, 3.418, 1.464, 0.244}.

- Correlation
- Correlation measures the statistical relationship between two attributes,particularly dependence of one attribute on another attribute.
- When two attributes are highly correlated with each other, they both vary at the same rate with each other either in the same or in opposite directions

Types of correlation



- For example, consider average temperature of the day and ice cream sales. Statistically, the two attributes that are correlated are dependent on each other and one may be used to predict the other. If there are sufficient data, future sales of ice cream can be predicted if the temperature forecast is known.
- correlation between two attributes does not imply causation, that is, one doesn’t necessarily cause the other. The ice cream sales and the shark attacks are correlated, however there is no causation.

- Both ice cream sales and shark attacks are influenced by the third attribute—the summer season.
- Generally, ice cream sales spikes as temperatures rise. As more people go to beaches during summer, encounters with sharks become more probable.
- Correlation between two attributes is commonly measured by the Pearson correlation coefficient (r),

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

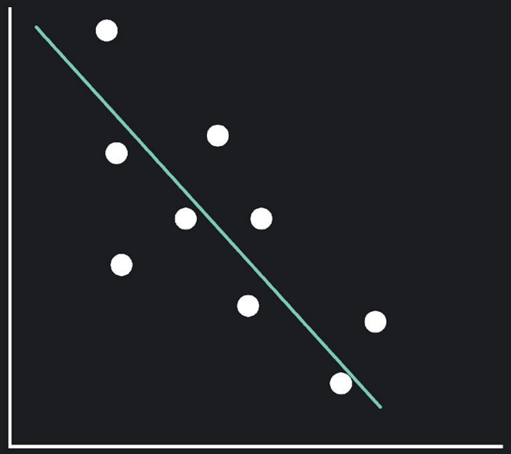
Where,

r = Pearson Correlation Coefficient

x_i = x variable samples $\quad$ y_i = y variable sample

$\bar{x}$ = mean of values in x variable $\quad$ $\bar{y}$ =mean of values in y variable

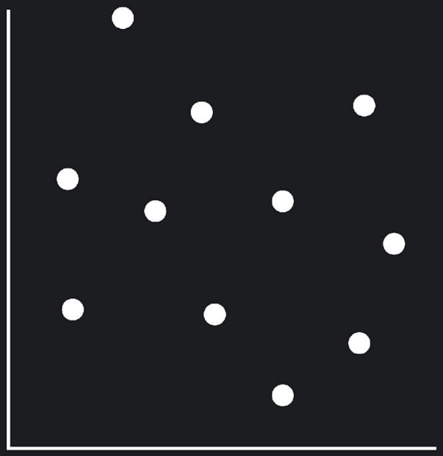
Negative Correlation



The plotted points are clustered around a downward sloping line.

As the values one variable decreases, the other tend to increase and vice versa.

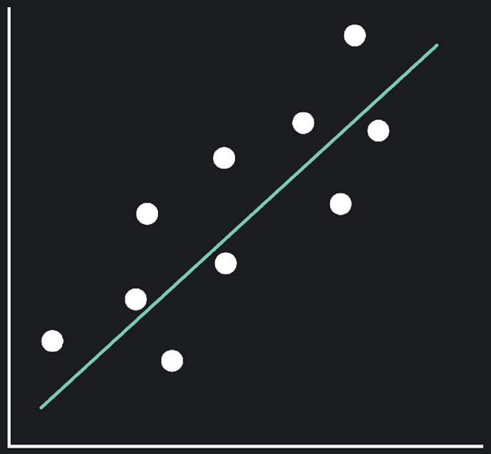
No Correlation



There is no linear trend.

There is no linear correlation between the two variables.

Positive Correlation



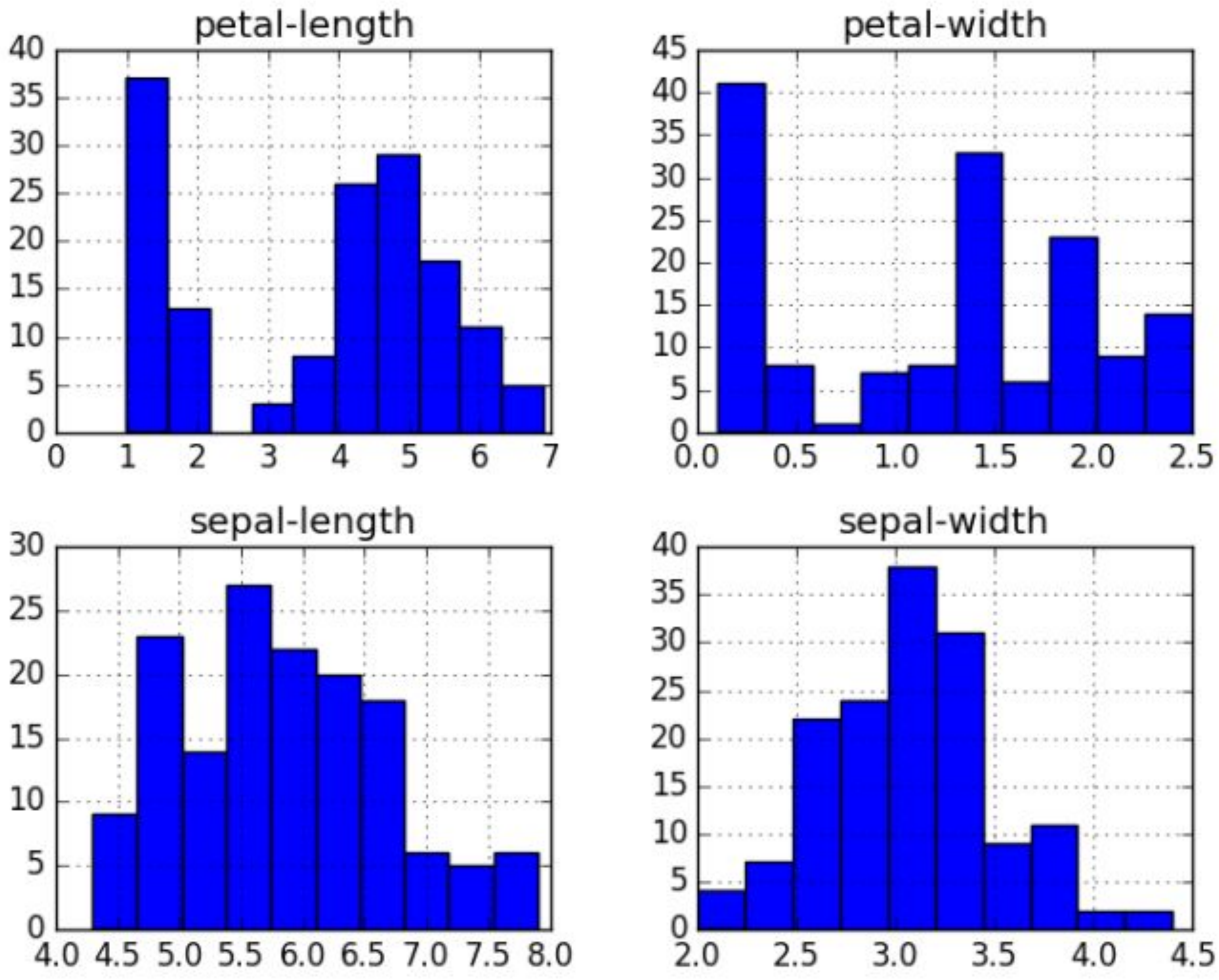
The plotted points are clustered around an upward sloping line.

As the values of one variable increase, the value of the other variable tends to increase as well.
As the value of one variable decreases, the value of the other also decreases.

3.4
DATA VISUALIZATION

- Visualizing data is one of the most important techniques of data discovery and exploration
- The discipline of data visualization enc**ompasses the methods of expressing data in an abstract visual form.** The visual representation of data provides easy comprehension of complex data with multiple attributes and their underlying relationships.
- **3.4.1 Univariate visualization**
- Visual exploration starts **with investigating one attribute at a time using univariate charts.** The techniques discussed in this section give an idea of how the attribute values are distributed and the shape of the distribution.

- 1.Histogram
- A histogram is one of the most basic visualization techniques to understand the frequency of the occurrence of values.
- It shows the distribution of the data by plotting the frequency of occurrence in a range. In a histogram, the attribute under inquiry is shown on the horizontal axis and the frequency of occurrence is on the vertical axis.


petal-length
40
35
30
25
20
15
10
5
0
0
1
2
3
4
5
6
7
petal-width
45
40
35
30
25
20
15
10
5
0
0.0
0.5
1.0
1.5
2.0
2.5
sepal-length
30
25
20
15
10
5
0
4.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
sepal-width
40
35
30
25
20
15
10
5
0
2.0
2.5
3.0
3.5
4.0
4.5

Histogram plot

- The towers or bars of a histogram are called bins. The height of each bin shows how many values from that data fall into that range.
- Width of each bin is = (max value of data – min value of data) / total number of bins
- The default value of the number of bins to be created in a histogram is 10
- A simple method to work our **how many bins** are suitable is to take the **square root of the total number of values** in your distribution.

- **2.Quartile (Box Plot)**

  Box plot is also known as a whisker plot, box-and-whisker plot, or simply a box-and whisker diagram. Box plot is a graphical representation of the distribution of a dataset.

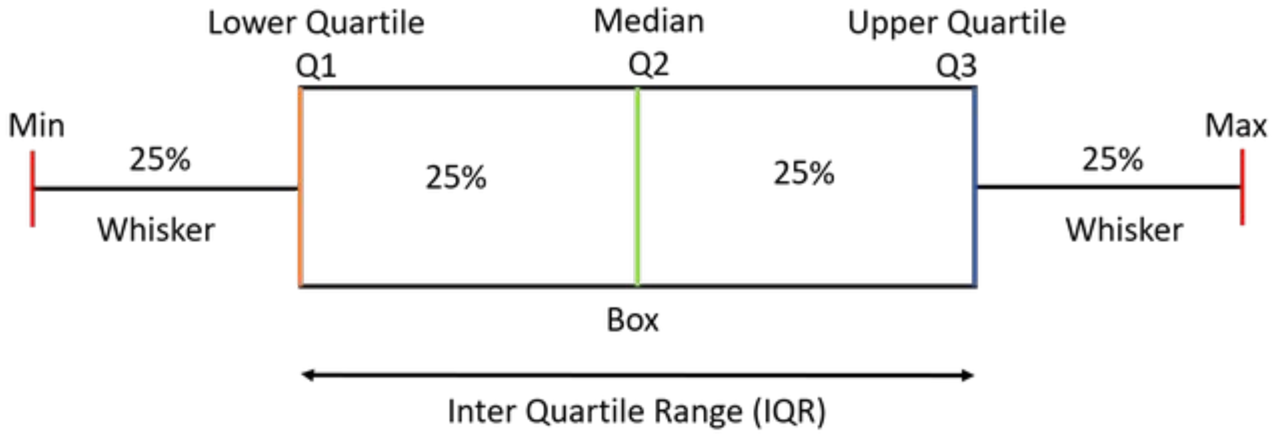
- A box whisker plot is a simple visual way of **showing the distribution of a continuous variable with information such as quartiles, median, and outliers,overlaid by mean and standard deviation.**

- The main attraction of box whisker or quartile charts is that distributions of multiple attributes can be compared side by side and the overlap between them can be deduced.

Elements of Box Plot

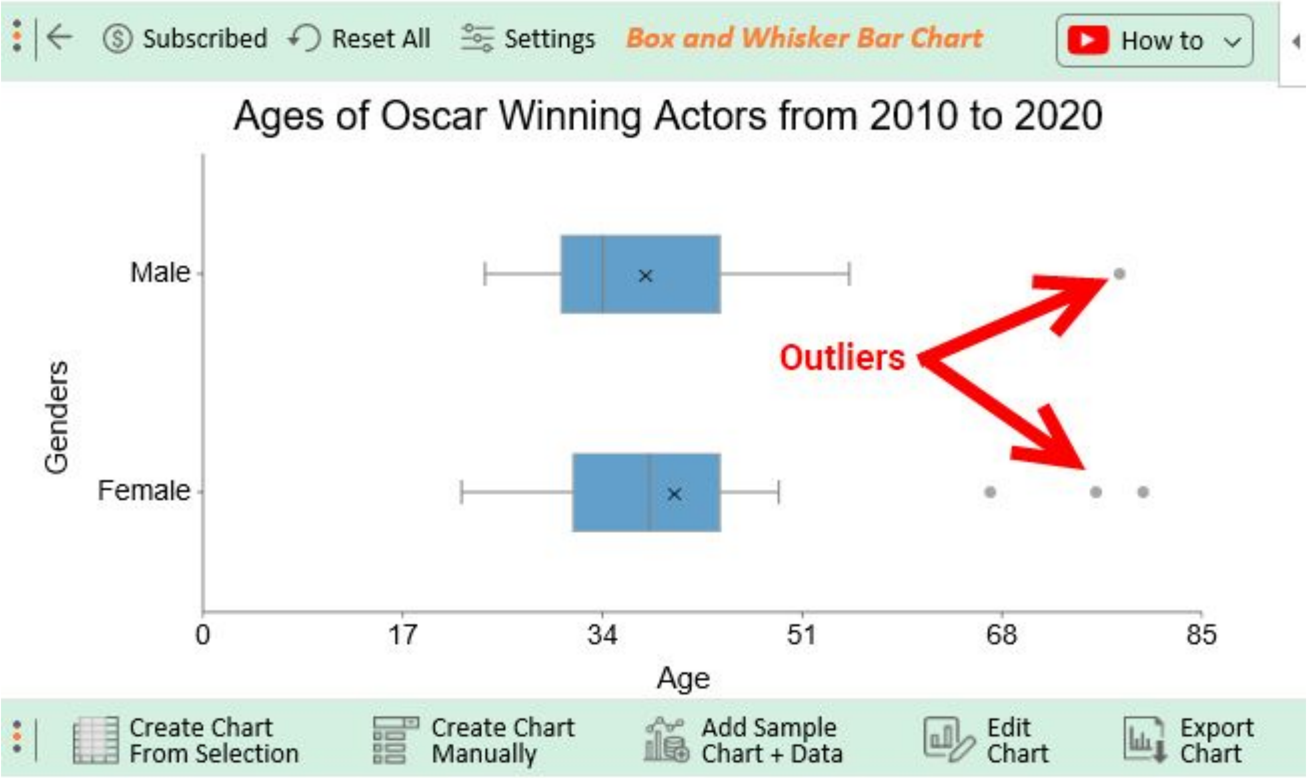
- A box plot gives a five-number summary of a set of data which is-
- **Minimum** – It is the minimum value in the dataset excluding the outliers.
- **First Quartile (Q1)** – 25% of the data lies below the First (lower) Quartile.
- **Median (Q2)** – It is the mid-point of the dataset. Half of the values lie below it and half above.
- **Third Quartile (Q3)** – 75% of the data lies below the Third (Upper) Quartile.
- **Maximum** – It is the maximum value in the dataset excluding the outliers.

Box Plot


Lower Quartile
Q1
Median
Q2
Upper Quartile
Q3
Min
25%
Whisker
25%
25%
Max
25%
Whisker
Box
Inter Quartile Range (IQR)

- The area inside the box (50% of the data) is known as the Inter Quartile Range. The IQR is calculated as
- **IQR = Q3-Q1**
- **Outlies are the data points below and above the lower and upper limit.**

Outlier representation in box plot

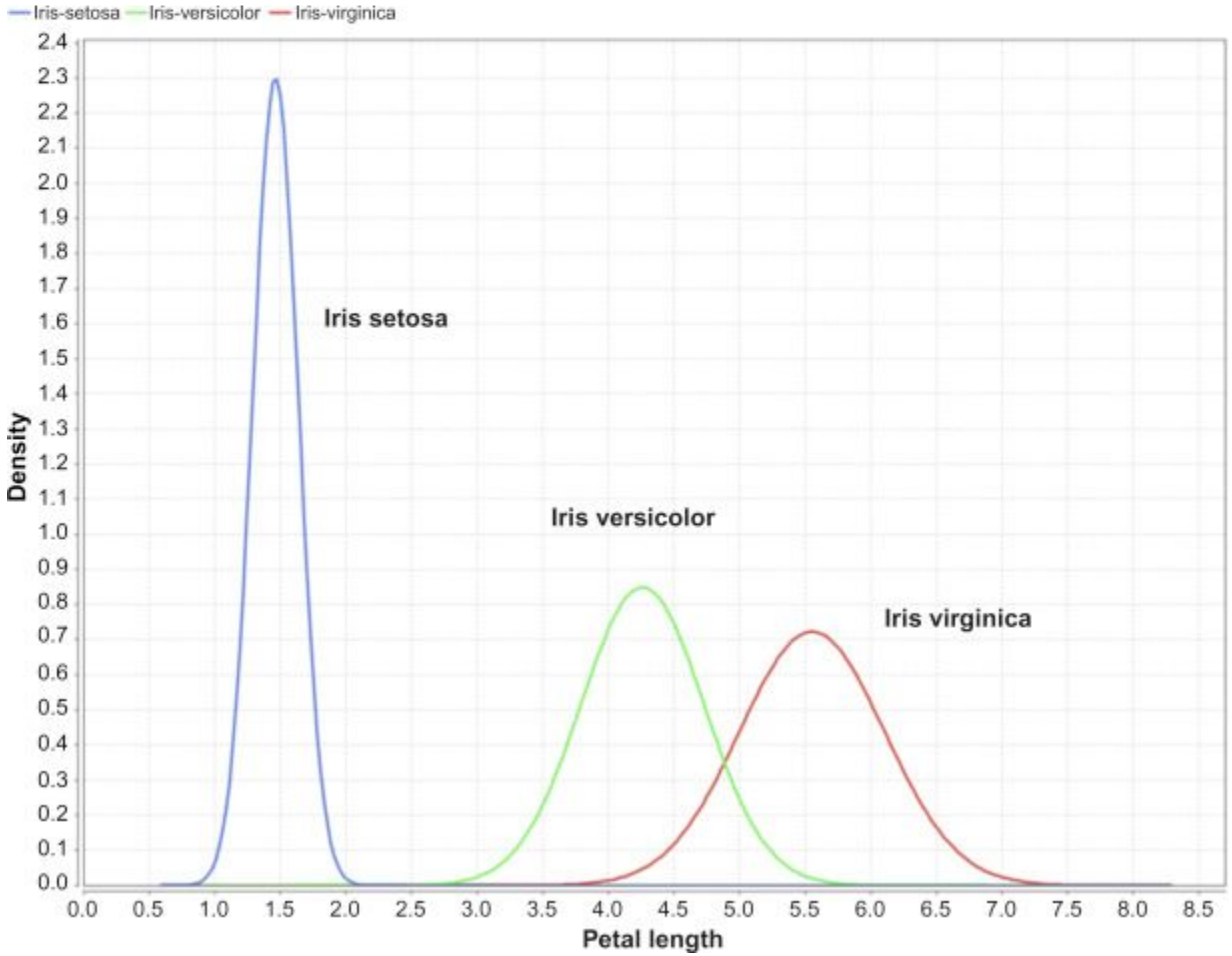


- **3.Distribution Chart**
- For continuous numeric attributes like petal length, **instead of visualizing the actual data in the sample, its normal distribution function can be visualized instead.** The normal distribution function of a continuous random variable is given by the formula:

$$f(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2} \qquad \text{for } -\infty < x < \infty$$

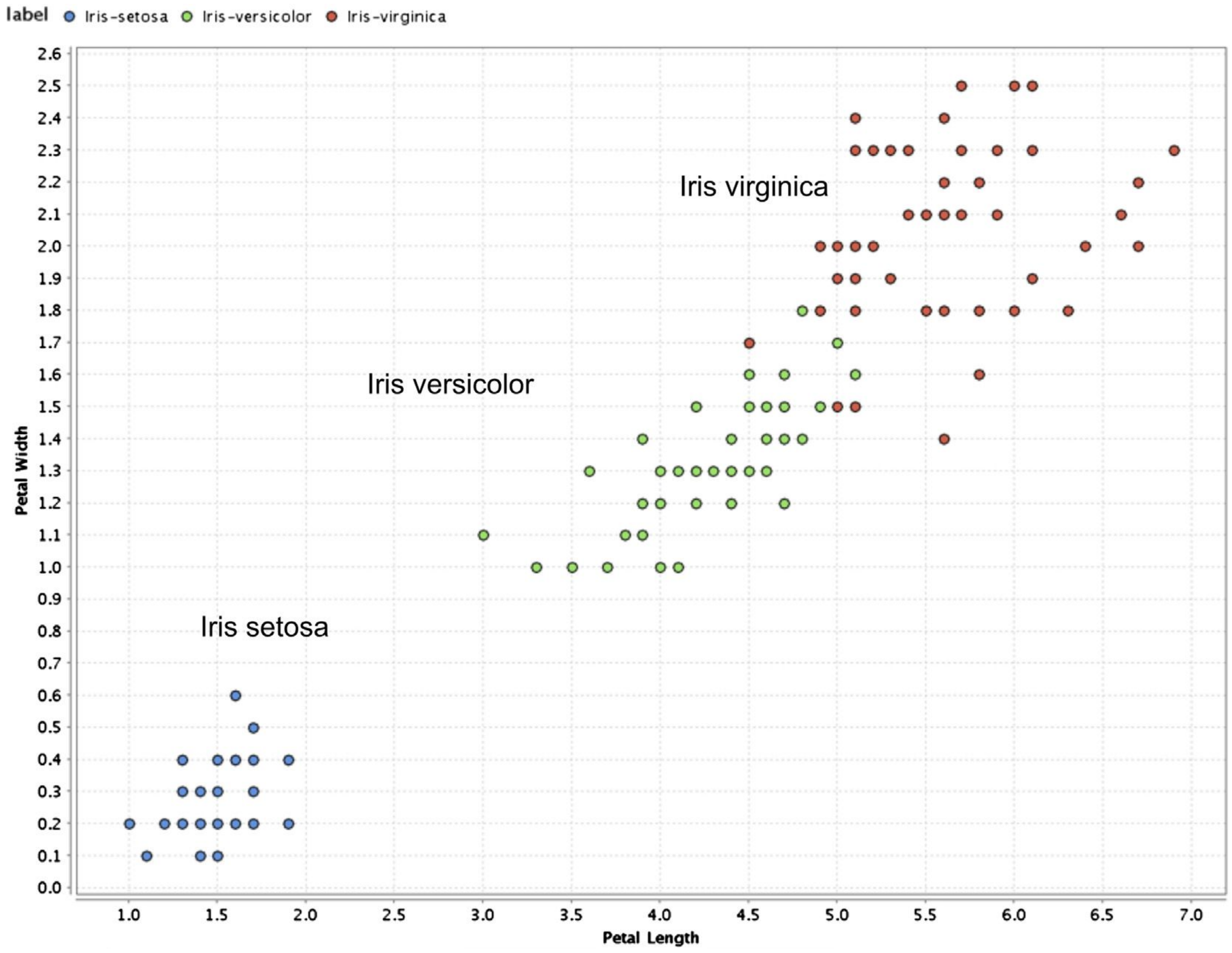
- where μ is the mean of the distribution and σ is the standard deviation of the distribution.

Distribution chart



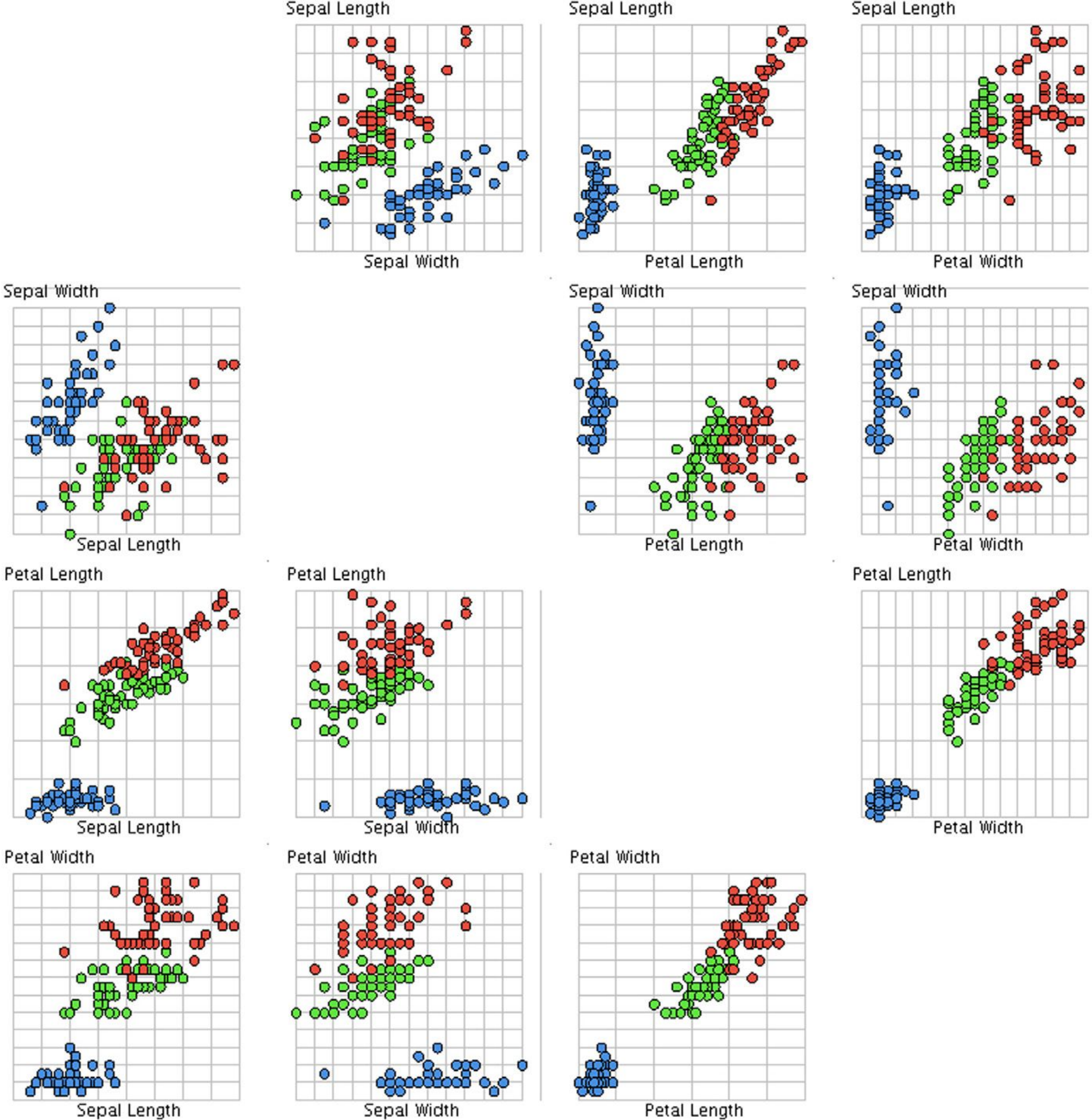
Multivariate Visualization

- 1.**Scatter plot**
- A scatter plot is one of the most powerful yet simple visual plots available. **In a scatter plot, the data points are marked in Cartesian space with attributes of the dataset aligned with the coordinates. The attributes are usually of continuous data type.**


label
Iris-setosa
Iris-versicolor
Iris-virginica
Iris virginica
Iris versicolor
Iris setosa
Petal Width
Petal Length
2.6
2.5
2.4
2.3
2.2
2.1
2.0
1.9
1.8
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
6.5
7.0

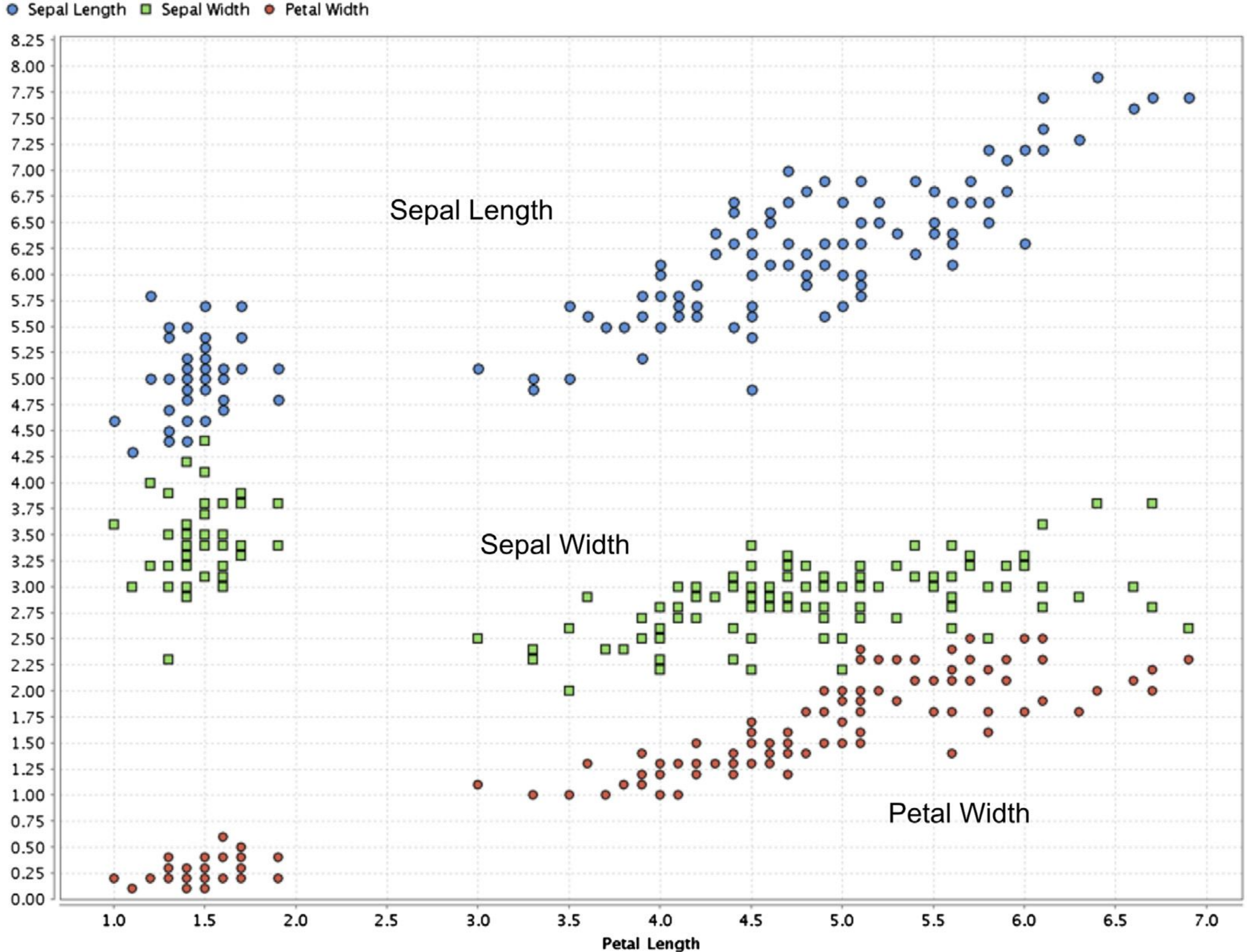
- The primary attribute is used for the x-axis coordinate. The secondary axis is shared with more attributes or dimensions.
- In this example (Fig. 3.11), the values on the y-axis are shared between sepal length, sepal width, and petal width. The name of the attribute is conveyed by colors used in data markers.
- Here, **sepal length is represented by data points occupying the topmost part of the chart, sepal width occupies the middle portion, and petal width is in the bottom portion**

- **2.Scatter Matrix**
- If the dataset has more than two attributes, it is important to look at combinations of all the attributes through a scatter plot. **A scatter matrix solves this need by comparing all combinations of attributes with individual scatter plots and arranging these plots in a matrix.**
-


Sepal Length
Sepal Width
Sepal Length
Petal Length
Sepal Length
Petal Width
Sepal Width
Sepal Length
Sepal Width
Petal Length
Sepal Width
Petal Width
Petal Length
Sepal Length
Petal Length
Sepal Width
Petal Length
Petal Width
Petal Width
Sepal Length
Petal Width
Sepal Width
Petal Width
Petal Length

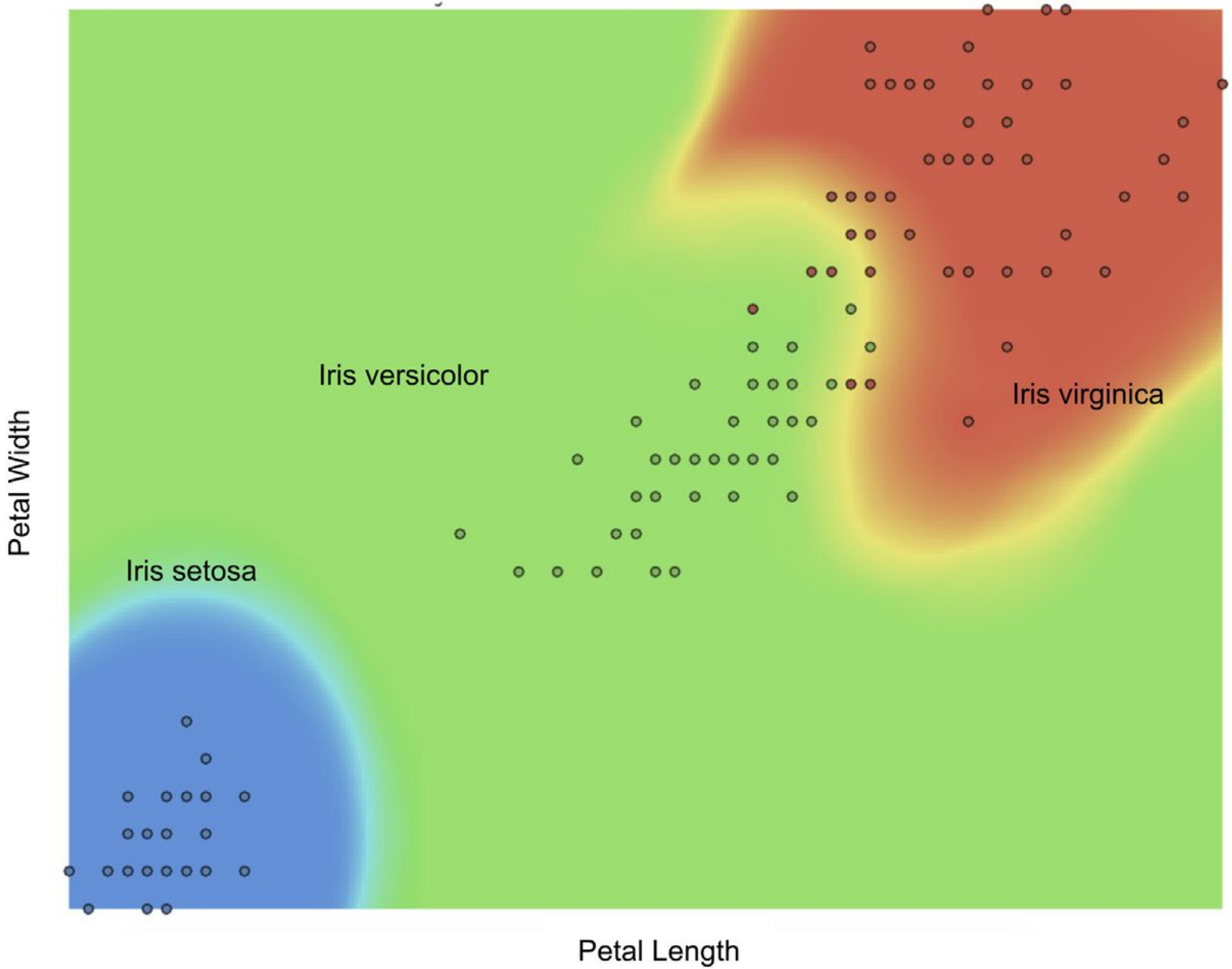
Scatter Multiple

- A scatter multiple is an enhanced form of a simple scatterplot where more than two dimensions can be included in the chart and studied simultaneously. The primary attribute is used for the x-axis coordinate. The secondary axis is shared with more attributes or dimensions.
- In this example (Fig. 3.11), the values on the y-axis are shared between sepal length, sepal width, and petal width. The name of the attribute is conveyed by colors used in data markers. Here, sepal length is represented by data points occupying the topmost part of the chart, sepal width occupies the middle portion, and petal width is in the bottom portion.
- Note that the data points are duplicated for each attribute in the y-axis. Data points are color-coded for each dimension in y-axis while the x-axis is anchored with one attribute petal length. All the attributes sharing the y-axis should be of the same unit or normalized.

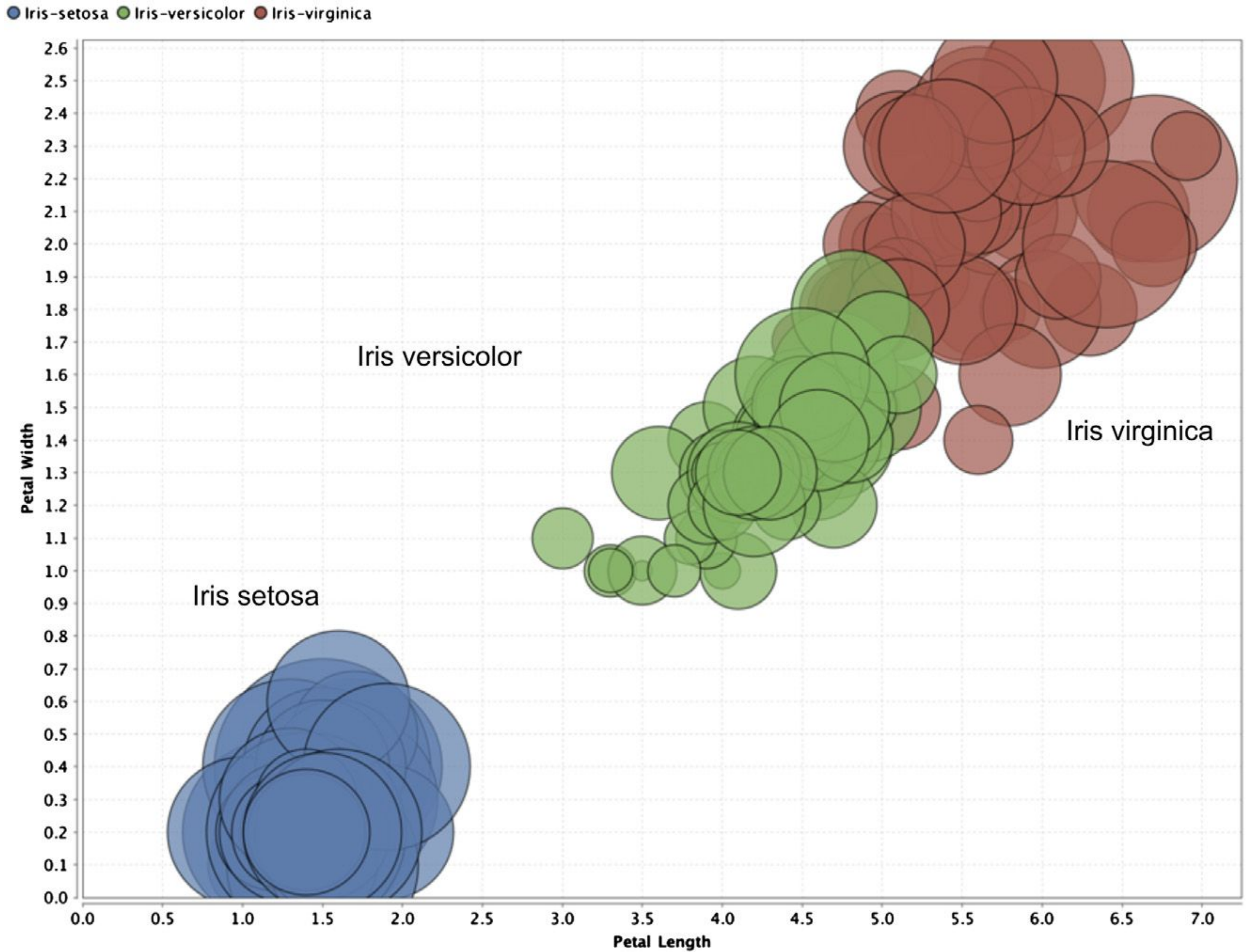


Sepal Length
Sepal Width
Petal Width
Sepal Length
Sepal Width
Petal Width
Petal Length

- **3.Density Chart**
- Density charts are similar to the scatter plots, **with one more dimension included as a background color.** The data point can also be colored to visualize one dimension, and hence, a total of four dimensions can be visualized in a density chart. In the example in Fig. **petal length is used for the x-axis, sepal length for the y-axis, sepal width for the background color, and class label for the data point color.**


Iris versicolor
Iris virginica
Iris setosa
Petal Width
Petal Length

- Bubble Chart
- A bubble chart is a variation of a **simple scatter plot with the addition of one more attribute, which is used to determine the size of the data point.**
- In the Iris dataset, **petal length and petal width are used for x and y-axis, respectively and sepal width is used for the size of the data point. The color of the data point represents a species class label**



Iris-setosa
Iris-versicolor
Iris-virginica
Petal Width
Petal Length
Iris versicolor
Iris setosa
Iris virginica
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
1.1
1.2
1.3
1.4
1.5
1.6
1.7
1.8
1.9
2.0
2.1
2.2
2.3
2.4
2.5
2.6
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
6.5
7.0

Associated Fields in Data Science

- **Descriptive statistics:** Computing mean, standard deviation, correlation,and other descriptive statistics, quantify the aggregate structure of a dataset. This is essential information for understanding any dataset in order to understand the structure of the data and the relationships within the dataset. They are used in the exploration stage of the data science process.
- **Exploratory visualization**: The process of expressing data in visual coordinates enables users to find patterns and relationships in the data and to comprehend large datasets.

- **Dimensional slicing**: Online analytical processing (OLAP) applications,which are prevalent in organizations, mainly provide information on the data through dimensional slicing, filtering, and pivoting.
- OLAP analysis is enabled by a unique database schema design where the data are organized as dimensions (e.g., products, regions, dates) and quantitative facts or measures (e.g., revenue, quantity). With a well defined database structure, it is easy to slice the yearly revenue by products or combination of region and products. These techniques are extremely useful and may unveil patterns in data

- **Hypothesis testing**: In confirmatory collected to evaluate whether a hypothesis has enough evidence to be supported or not.In general, data science is a process where many hypotheses are generated and tested based on observational data.
- **Data engineering:** Data engineering is the process of sourcing,organizing, assembling, storing, and distributing data for effective analysis and usage.Data engineering helps source and prepare for data science learning algorithms.

- **Business intelligence:** Business intelligence helps organizations consume data effectively. In Combination with data science, both the past and the predicted future data can be combined. BI can hold and distribute the results of data science.

- End of Module 1